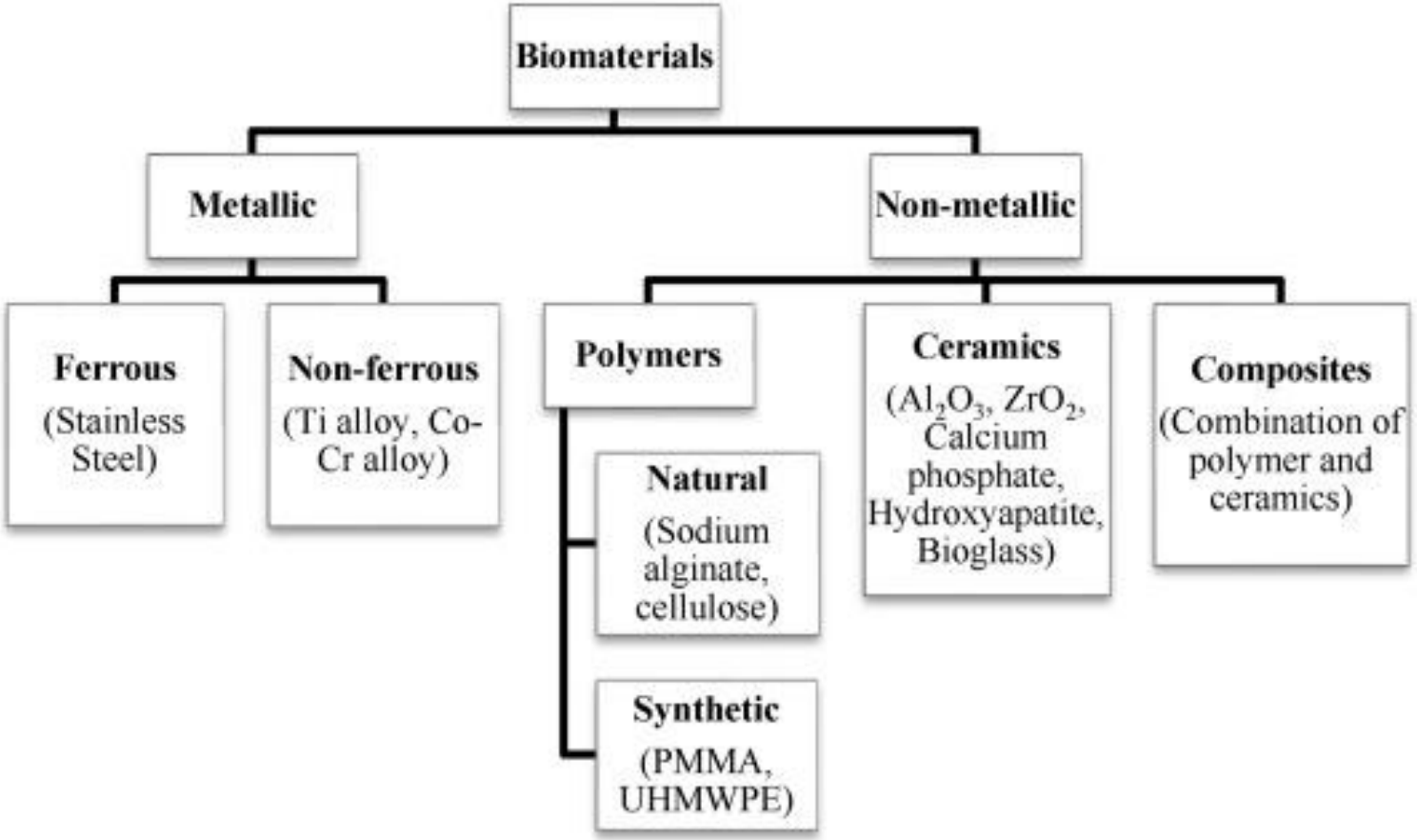
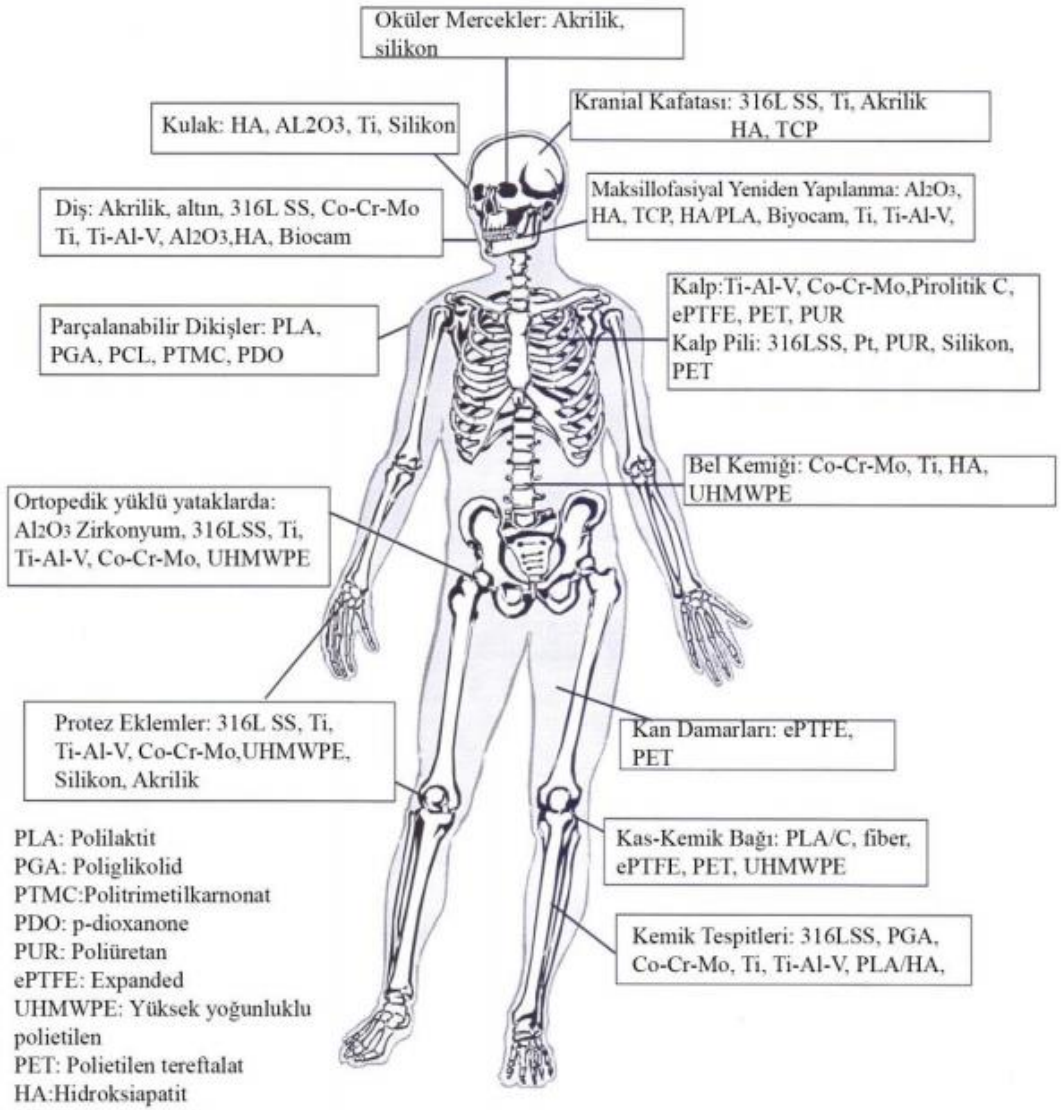
Biomedical Materials:
Ortheopedic implants (Hip Prothesis)

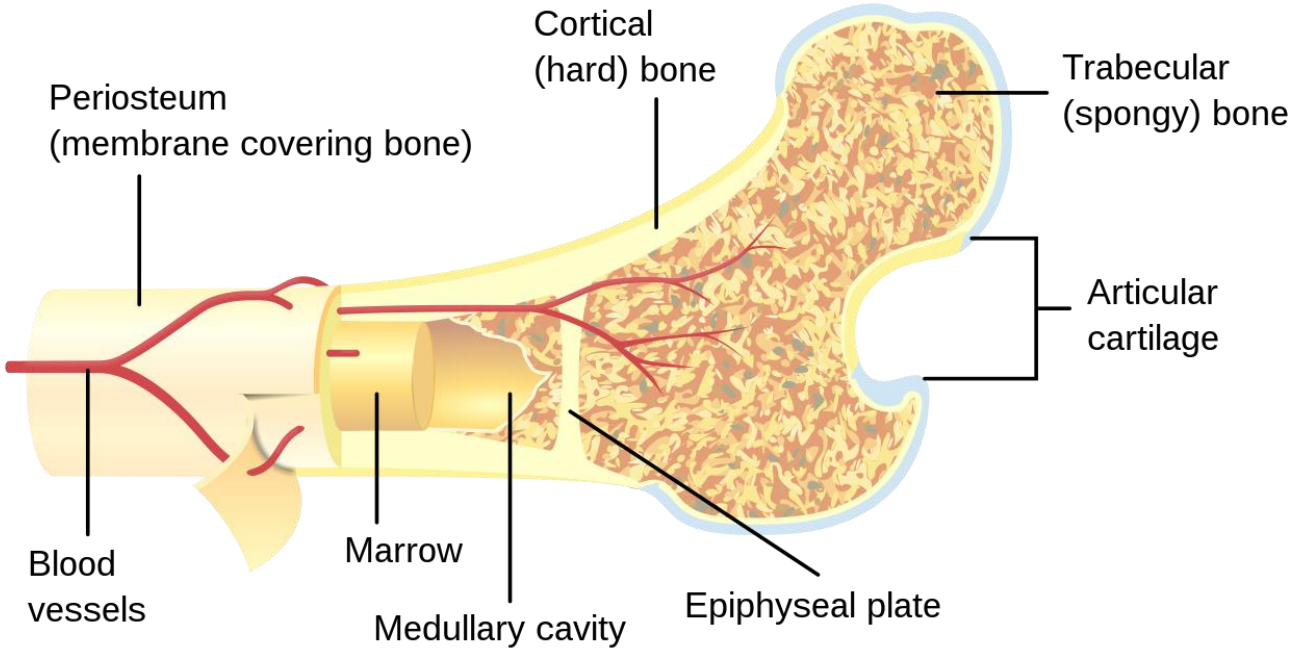
Assoc. Prof. Dr. Ali Can Zaman

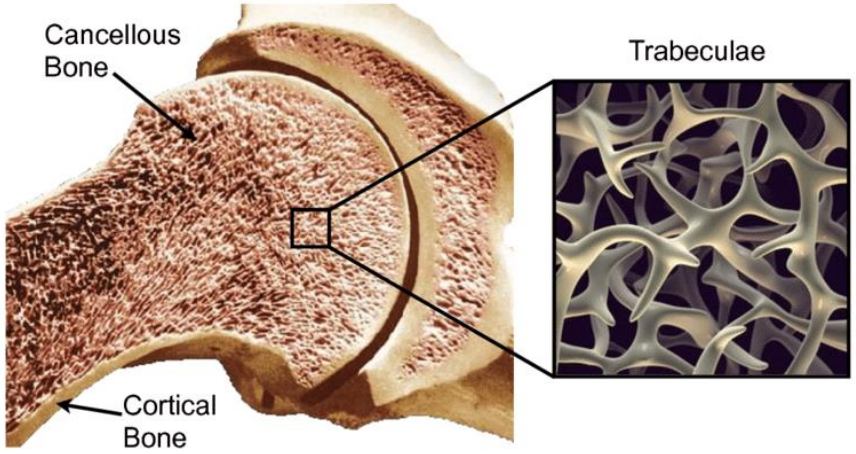
Classifiction of biomaterials




Oküler Mercekler: Akrilik, silikon
Kulak: HA, AL2O3, Ti, Silikon
Kranial Kafatası: 316L SS, Ti, Akrilik HA, TCP
Diş: Akrilik, altın, 316L SS, Co-Cr-Mo Ti, Ti-Al-V, Al2O3,HA, Biocam
Maksillofasiyal Yeniden Yapılanma: Al2O3, HA, TCP, HA/PLA, Biyocam, Ti, Ti-Al-V,
Kalp:Ti-Al-V, Co-Cr-Mo,Pirolitik C, ePTFE, PET, PUR
Kalp Pili: 316LSS, Pt, PUR, Silikon, PET
Parçalanabilir Dikişler: PLA, PGA, PCL, PTMC, PDO
Bel Kemiği: Co-Cr-Mo, Ti, HA, UHMWPE
Ortopedik yüklü yataklarda: Al2O3 Zirkonyum, 316LSS, Ti, Ti-Al-V, Co-Cr-Mo, UHMWPE
Protez Eklemler: 316L SS, Ti, Ti-Al-V, Co-Cr-Mo,UHMWPE, Silikon, Akrilik
Kan Damarları: ePTFE, PET
Kas-Kemik Bağı: PLA/C, fiber, ePTFE, PET, UHMWPE
Kemik Tespitleri: 316LSS, PGA, Co-Cr-Mo, Ti, Ti-Al-V, PLA/HA,
PLA: Polilaktit
PGA: Poliglikolid
PTMC:Politrimetilkarnonat
PDO: p-dioxanone
PUR: Poliüretan
ePTFE: Expanded
UHMWPE: Yüksek yoğunluklu polietilen
PET: Polietilen tereftalat
HA:Hidroksiapatit
SS: Paslanmaz Çelik

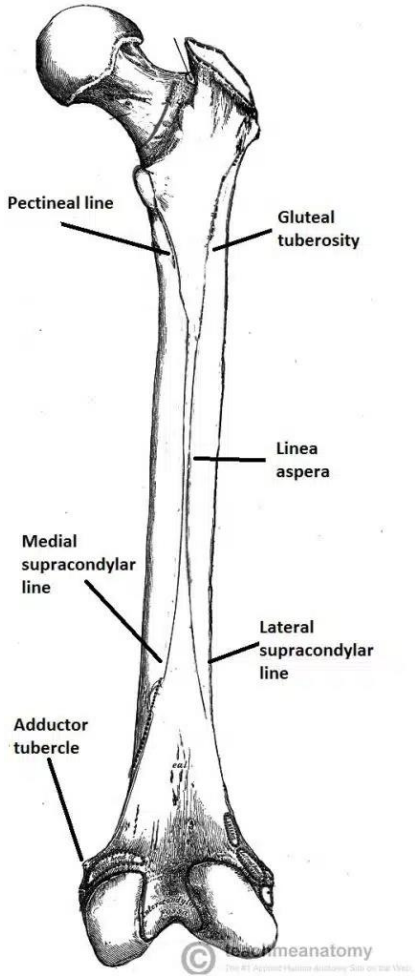
Bone tissue

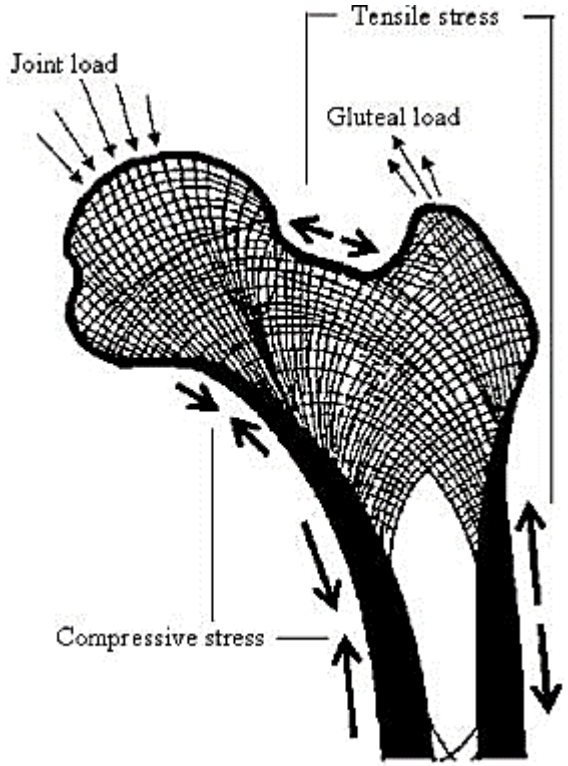


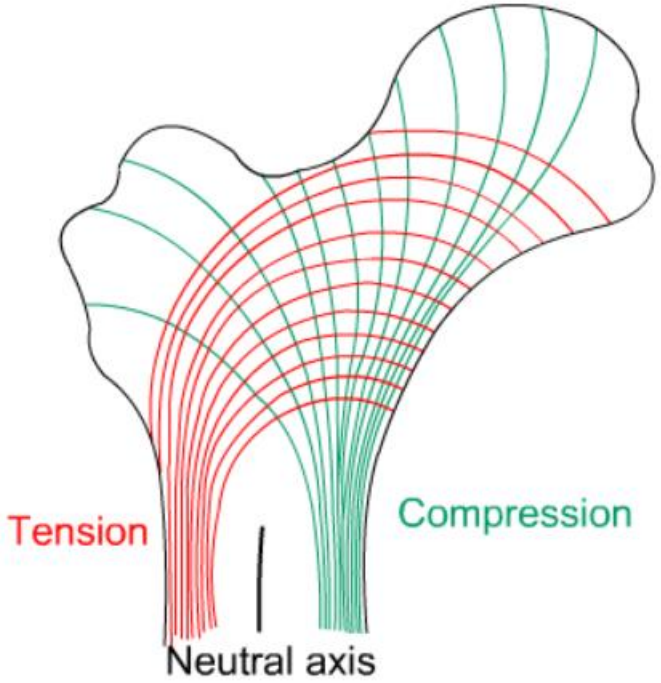


Cortical bone is highly dense and organized into a compact structure, with minimal spaces compared to spongy or cancellous bone (found inside bones).

Femur and distribution of stresses

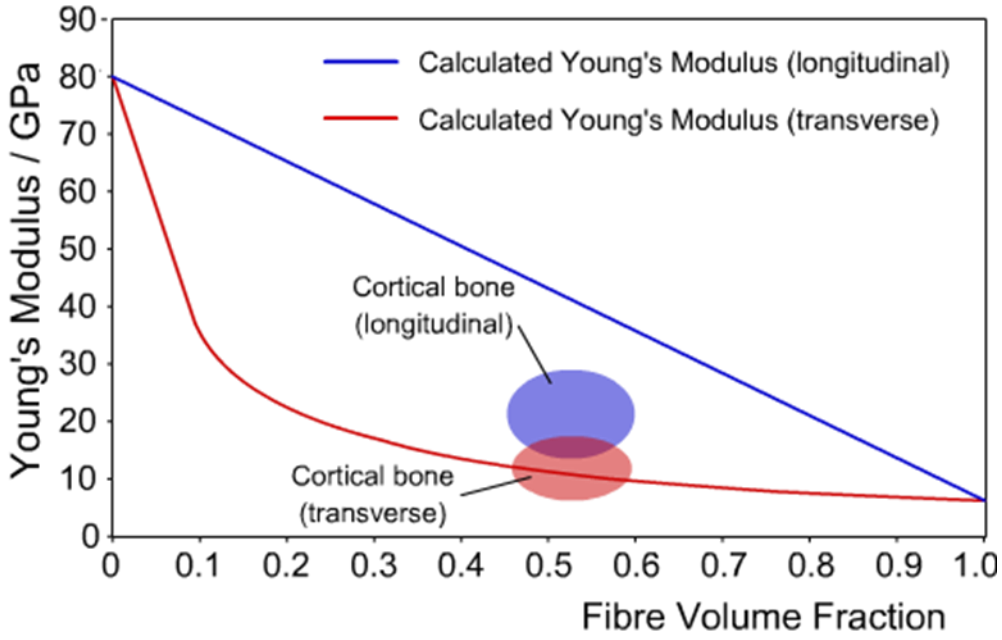






Mechanical properties of bone tissue

One can be considered a composite material consisting of collagen fibers and an inorganic matrix. The properties of a composite material (such as mechanical strength, hardness, density, etc.) can be calculated as a weighted average of the properties of its components. In the case of bone, the organic collagen provides flexibility and toughness, while the inorganic hydroxyapatite matrix contributes to stiffness and compressive strength.



In the longitudinal direction the difference is large, indicating the model does not give an accurate picture of the behaviour of bone. This difference occurs because the composite model of the microstructure of bone is highly simplified, since the collagen fibres are not aligned parallel to the axis of the osteons, and the bone mineral exists as discrete crystals, rather than forming a continuous matrix.

Mechanical properties of bone tissue

This table presents the **Young's Modulus (E)** for various materials relevant to bone, including collagen, bone mineral (hydroxyapatite), and cortical bone in two different directions (longitudinal and transverse).

Material	Young's Modulus, E (GPa)
Collagen (dry)	6
Bone mineral (Hydroxyapatite)	80
Cortical bone, longitudinal	11-21
Cortical bone, transverse	5-13

This indicates that collagen is quite flexible and not very stiff, which is expected given its role in providing toughness and flexibility to bone.

Hydroxyapatite is responsible for the bone's ability to resist compressive forces.

Cortical bone (compact bone) is stiffer when measured along the longitudinal direction (parallel to the bone's length). This direction aligns with the bone's natural load-bearing axis.

This reflects the anisotropic nature of bone

Mechanical properties of bone tissue

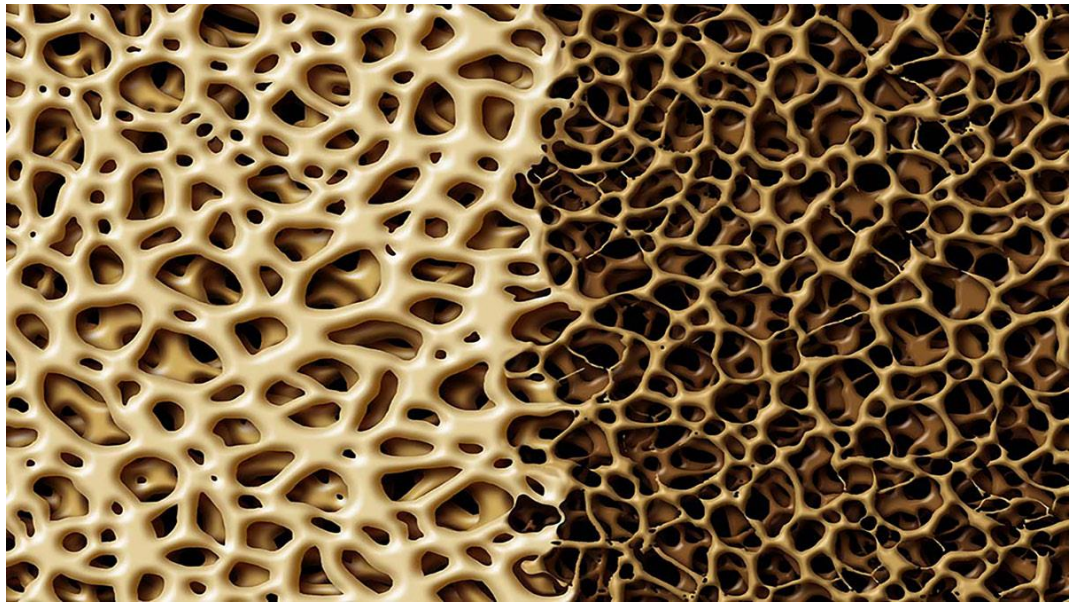
Different bones in the body are designed to withstand different forces depending on their location and function. For example, long bones like the femur and tibia, which bear most of the body's weight, are structured to resist high compressive forces, so they have high compressive strength.

	Longitudinal direction	Transverse direction
Tensile strength (MPa)	60-70	~50
Compressive strength (MPa)	70-280	~50

Bone remodelling

There is a continuous process in which bones are constantly broken down and rebuilt through the activity of two types of specialized cells: **osteoblasts** and **osteoclasts**.

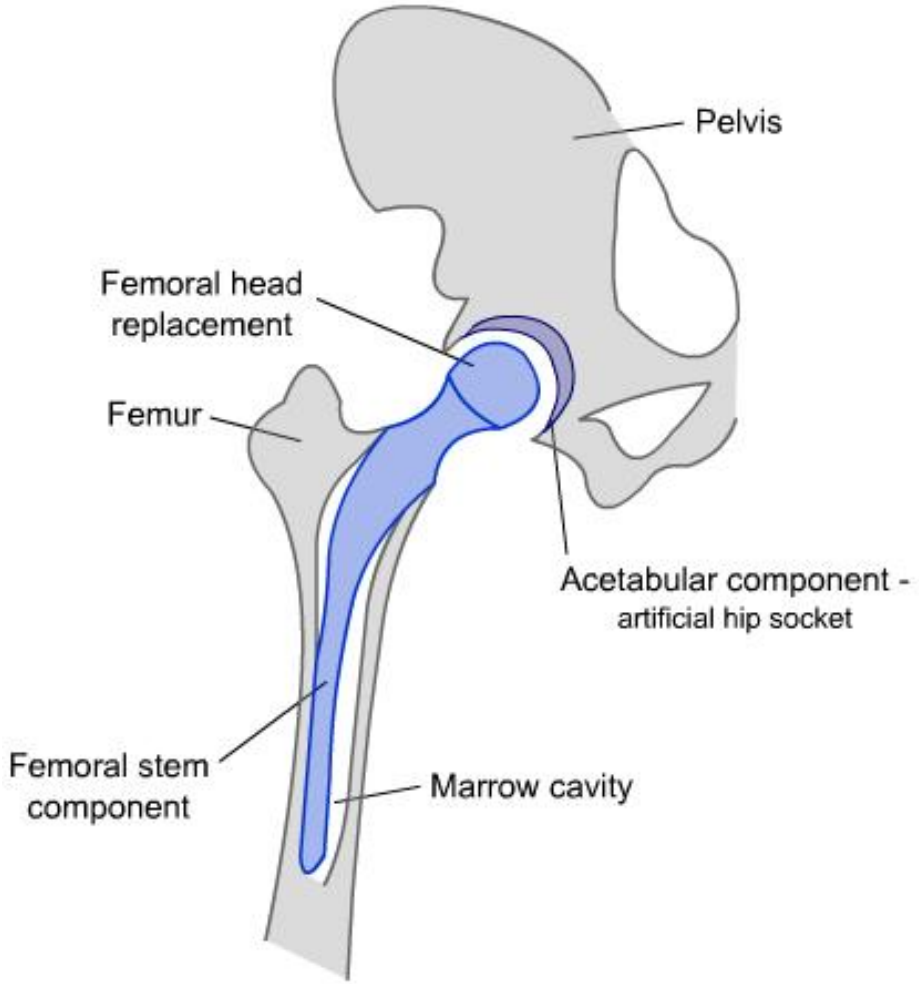
The combined activity of osteoblasts (building) and osteoclasts (breaking down) allows bones to adapt to mechanical stresses, **repair microdamage**, and maintain healthy bone structure throughout a person's life.





Osteoclasts are cells that break down bone tissue.

Osteoblasts are cells responsible for forming new bone tissue.





Femur with cemented titanium alloy prosthesis.
(S. Blatcher, PhD Thesis, Queen Mary College, London, 1995)

The hip prothesis

The **femoral stem** is inserted into the shaft of the femur, properly positioning the body of the implant. The femoral stem essentially replaces the bone within the femur and has a structural role that needs to be as close as possible to the strength and durability of natural bones.



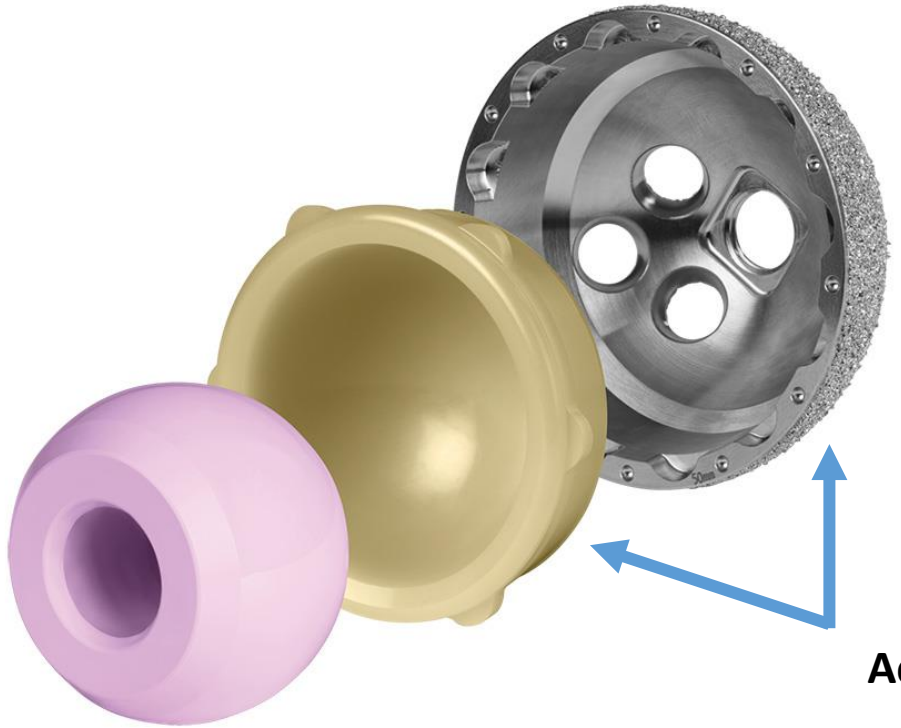
The hip prothesis

The **femoral head component** must meet certain requirements: it must have a specific shape that fits perfectly into the acetabular cup/hip socket, and its surface must have a low coefficient of friction to allow smooth movement.



The hip prothesis

The **acetabular cup** fits into the hip socket and takes over the role of the worn-out cartilage. As such, this component must have a low wear rate and, at the same time, minimize the wear of the femoral head component.



Acetabular cup

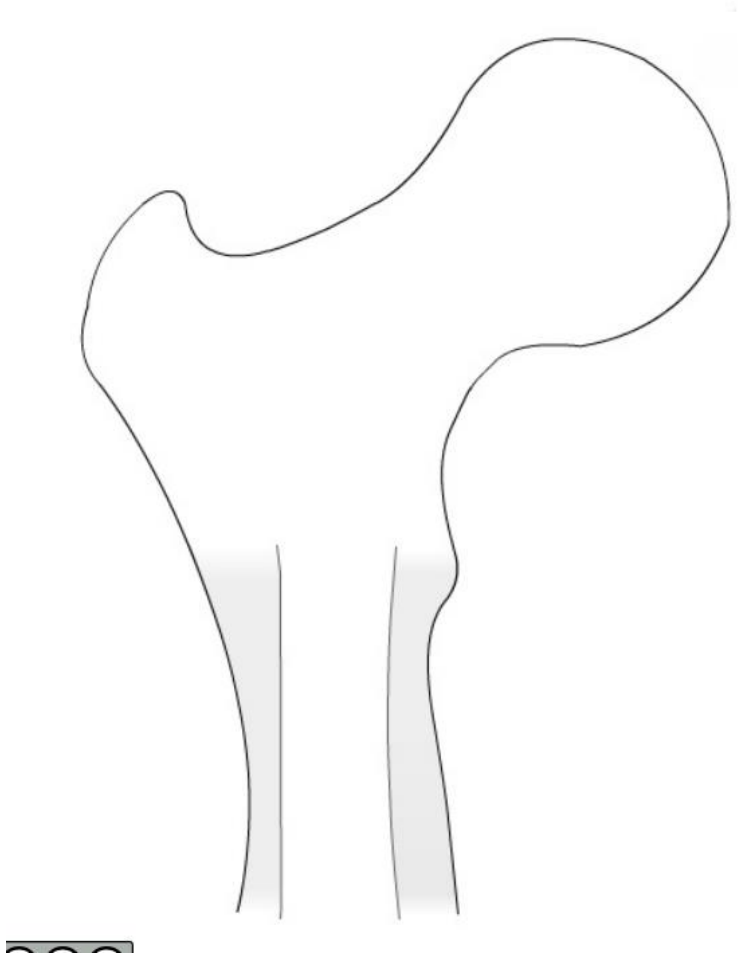
The hip prothesis

Biological Environment
The human body is not an easy environment for a material to function in for an extended period. The material must be able to operate for years at a temperature of 37°C in a highly humid and corrosive environment.

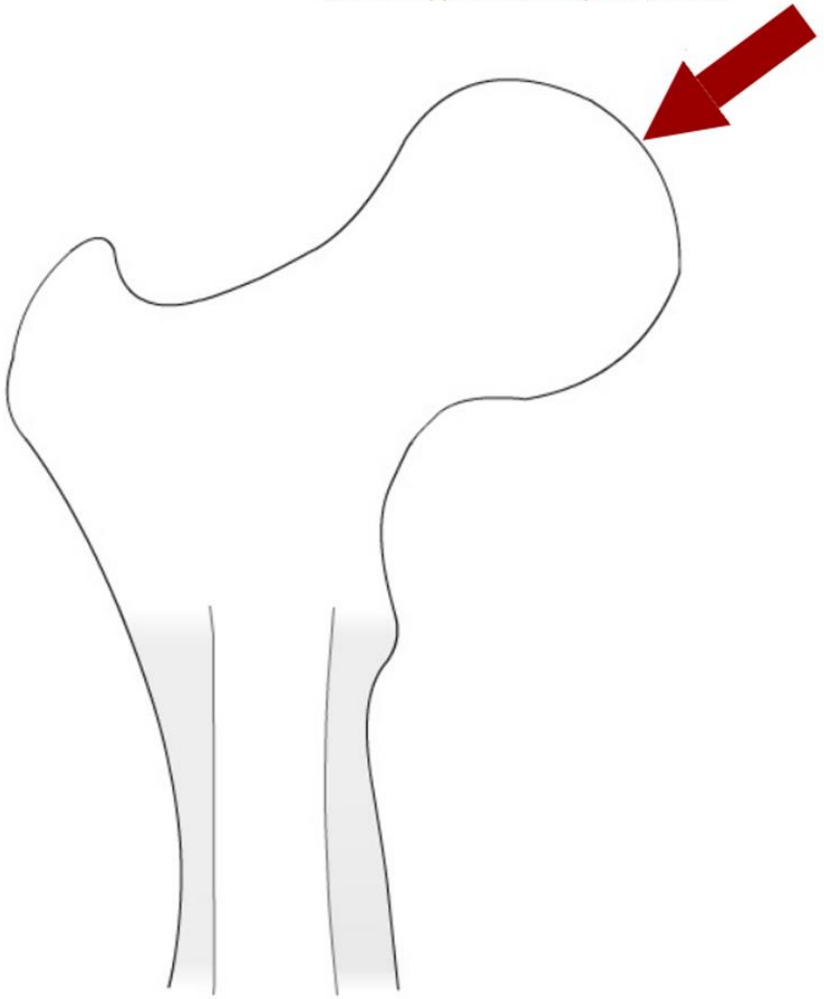
Tissue-Implant Interaction
Implant materials should be designed to minimize the adverse effects that a foreign substance introduced into the body might cause. The immune system typically attacks anything originating from outside the body, which can lead to inflammation. High levels of certain metals in the bloodstream can cause various issues, including cytotoxicity and cancer development. Therefore, it is crucial to choose materials that have minimal negative effects on the body.

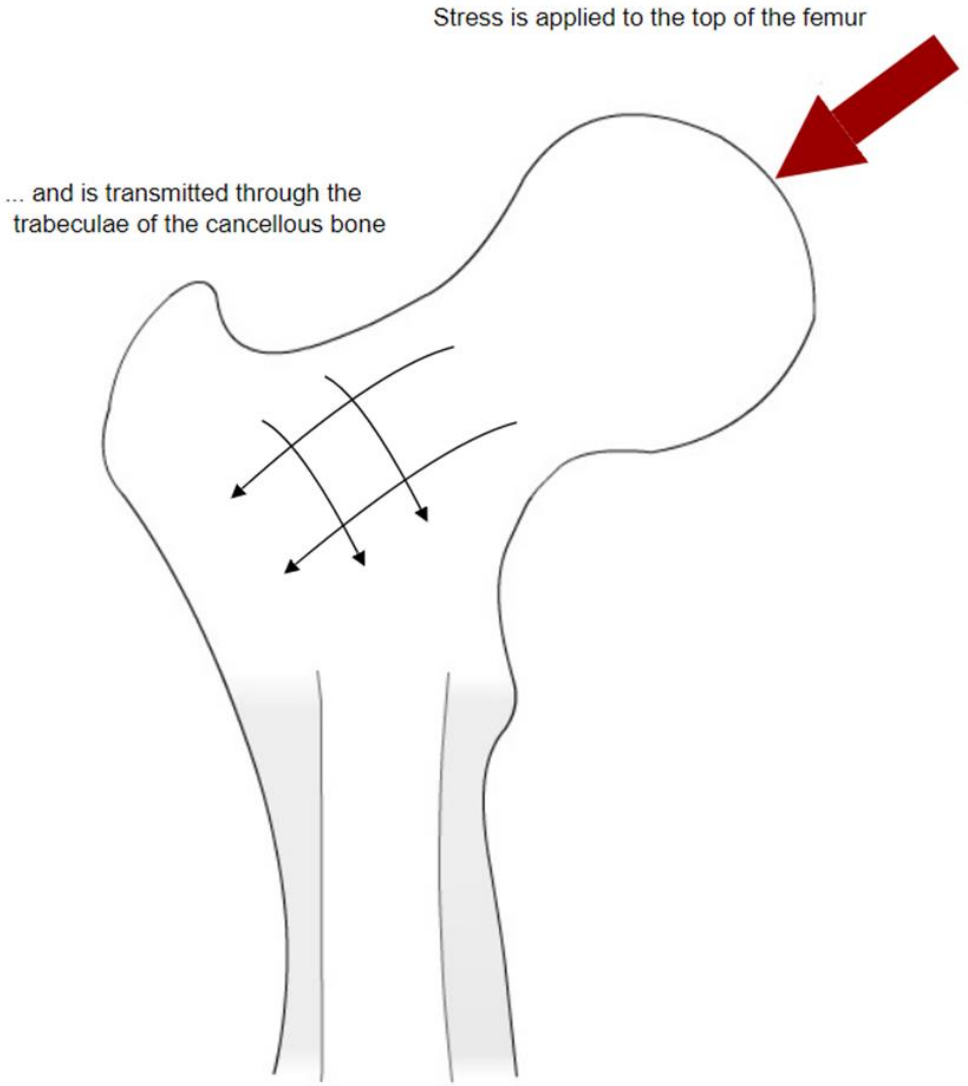
Cost
As with most areas in materials science, cost is a significant factor in material selection. Manufacturers often need to find an appropriate balance between the material's performance and its cost.

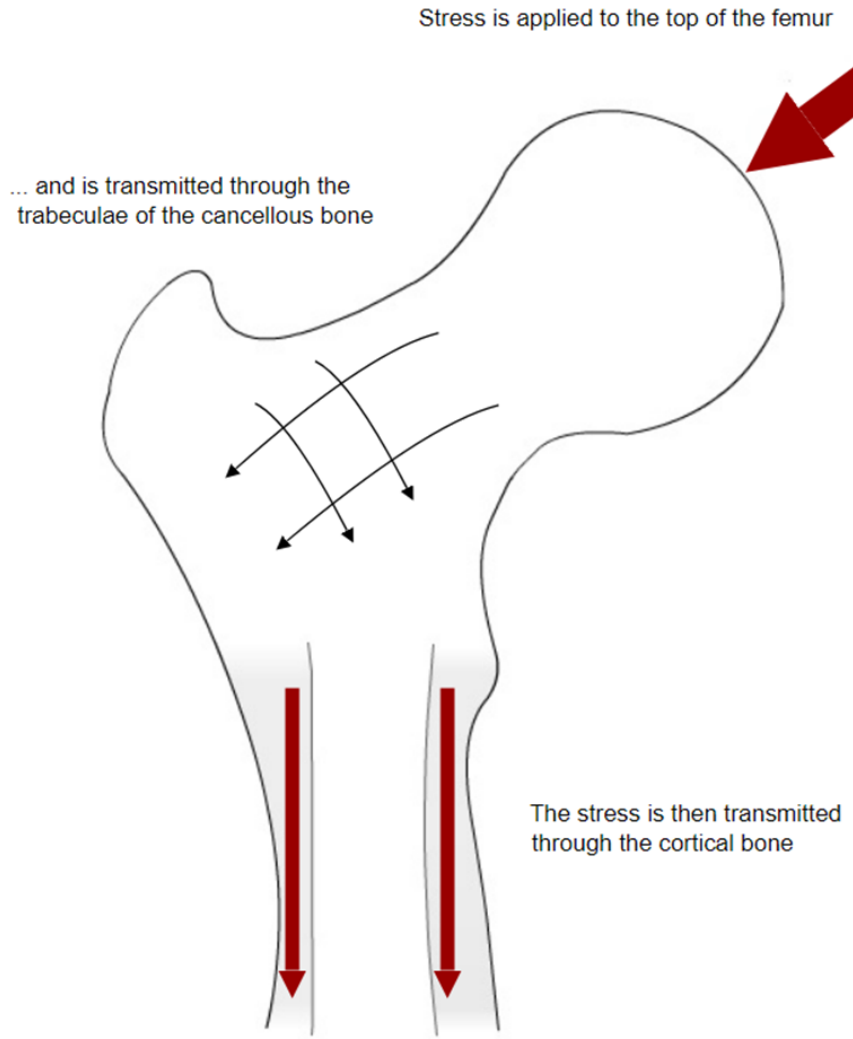
Materials selection

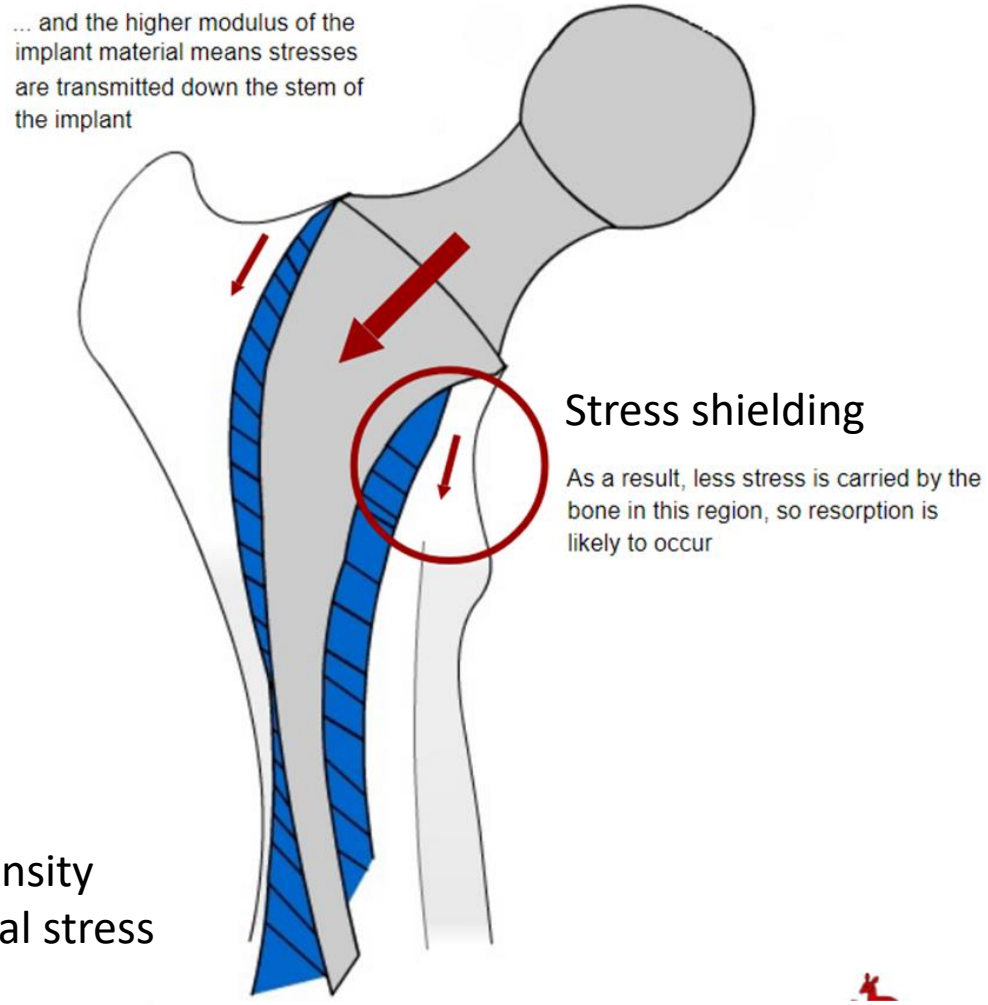


Stress is applied to the top of the femur




Stress is applied to the top of the femur
... and is transmitted through the trabeculae of the cancellous bone


Stress is applied to the top of the femur
... and is transmitted through the trabeculae of the cancellous bone
The stress is then transmitted through the cortical bone



Stress shielding is the reduction in bone density (osteopenia) as a result of removal of typical stress from the bone by an implant

Materials selection for hip replacement

Requirements from stem: mechnical properties, biocompatibility, longevity

Options:

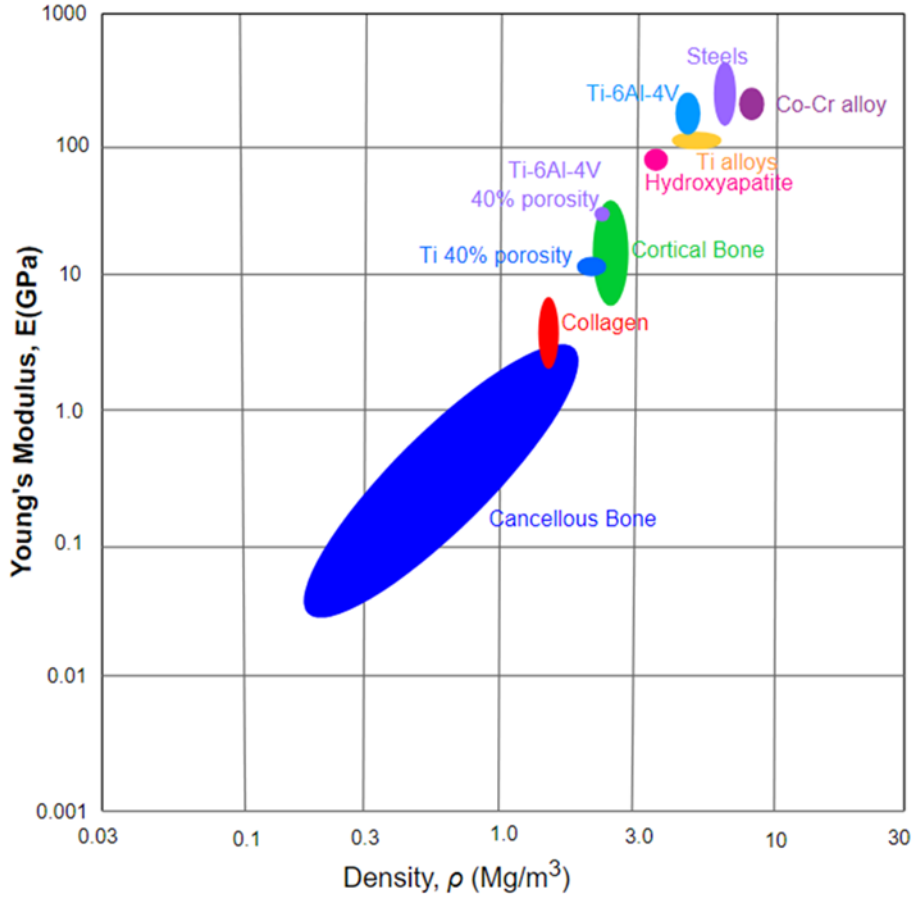
Polymers: low mechanical strength

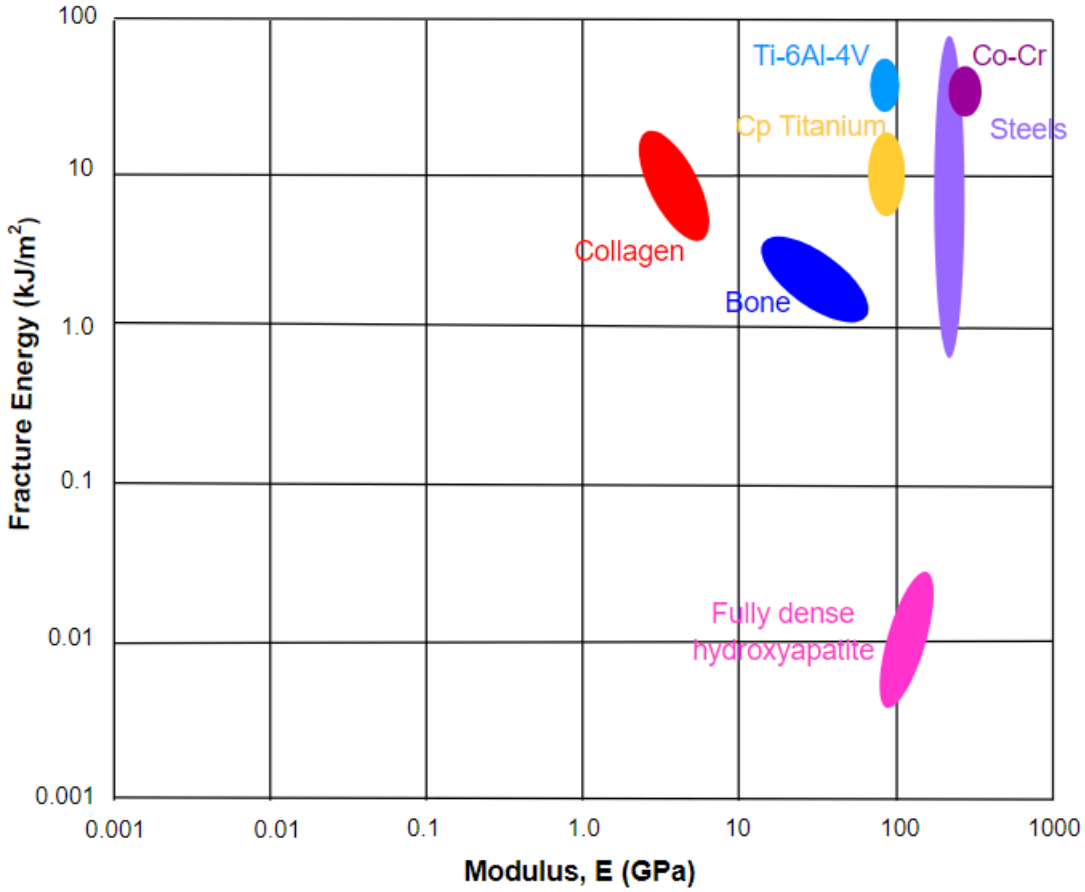
Metals: can have corrosion problems, but highmechanical strength

Ceramics: fragile

Elasticity modulus is critical for the material
that is to be selected for stem construction.

Materials selection for hip replacement





Materials selection for hip replacement

Stainless steel: Fe, Cr, Ni, Co alloy

High elasticity modulus of stainless steel accounts for stress shielding.

Nickel can be allergic

Cost: relatively low cost



Materials selection for hip replacement

Co-Cr alloys: Co-Cr-Mo

Cobalt-chromium-molybdenum (CoCrMo) alloy is one of the most used metals in total hip replacement (THR) due to the alloy's superior corrosion qualities and biocompatibility.

High elasticity modulus results in stress shielding

Production cost is higher than stainless steel



Materials selection for hip replacement

Titanium alloys

Titanium allys are **bioinert**. This property minimizes the risk of the body rejecting the implant

Titanium alloys have better **creep resistance** than traditional stainless steels

Titanium alloys are five Times More Expensive Per Kilogram Than Steel

Elasticity modulus (108 GPa) is half of steel: good for stress shielding but not enough

Titanium alloys exhibit **excellent corrosion resistance** due to the formation of a passive oxide layer (TiO_2) on their surface

Materials selection for hip replacement

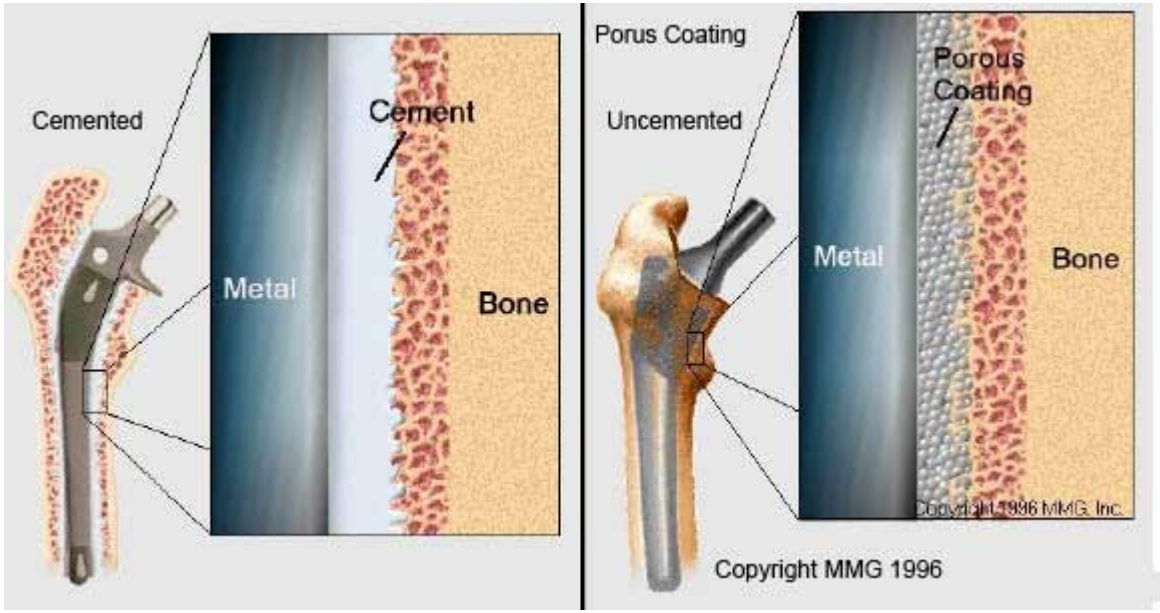
Ti-6Al-4V is chosen over commercially püre (CP) titanium are related to its **superior mechanical properties**, **biocompatibility**, and **performance** in load-bearing implants like hip replacements.

- Ti-6Al-4V (annealed) has a tensile strength of **~830-900 MPa**.
- CP titanium (Grade 2) has a tensile strength of **~300-500 MPa**.

Ti-6Al-4V is easier to work with in advanced manufacturing techniques such as **additive manufacturing (3D printing)**, **casting**, and **machining**.

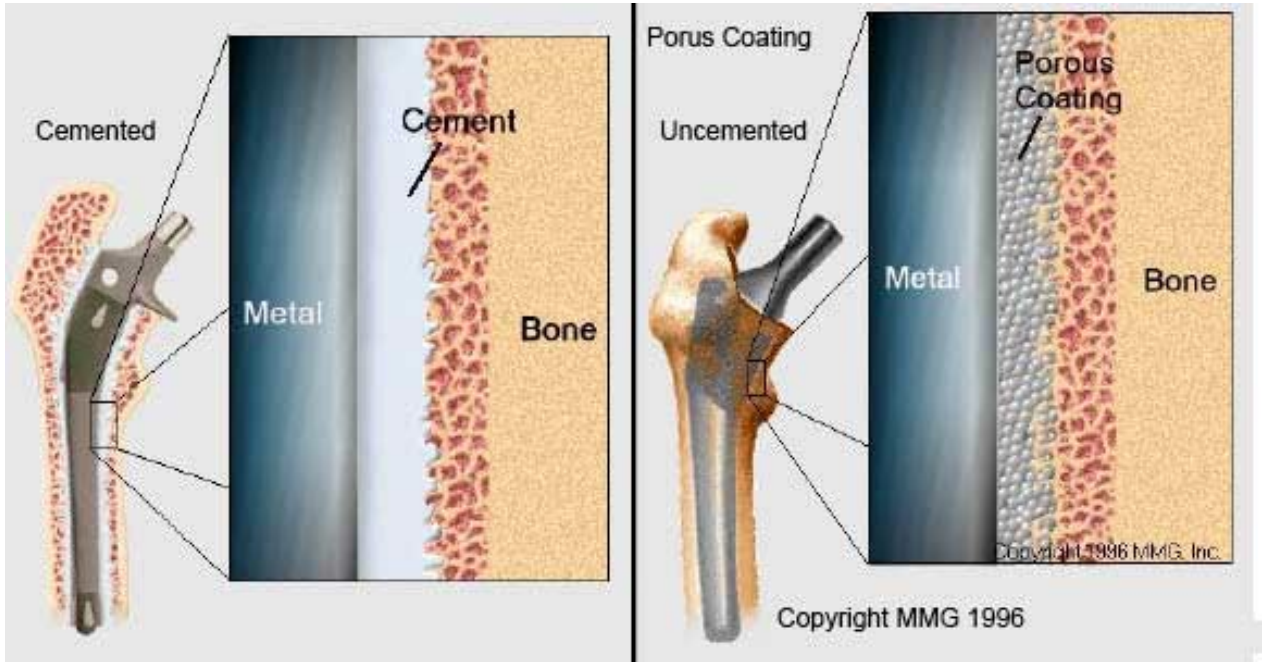
Femoral stem fixation

Polymethylmethacrylate (PMMA) is widely used as bone cement to fix the femoral stem to the bone in total hip replacements. PMMA works by filling the space between the implant and the bone, effectively securing the implant in place. During the curing process (polymerization of PMMA), the material releases heat (**exothermic reaction**). This **heat can cause cell death** (thermal necrosis) in the surrounding bone tissue, which is a recognized drawback of using PMMA bone cement.



Alternative Fixation

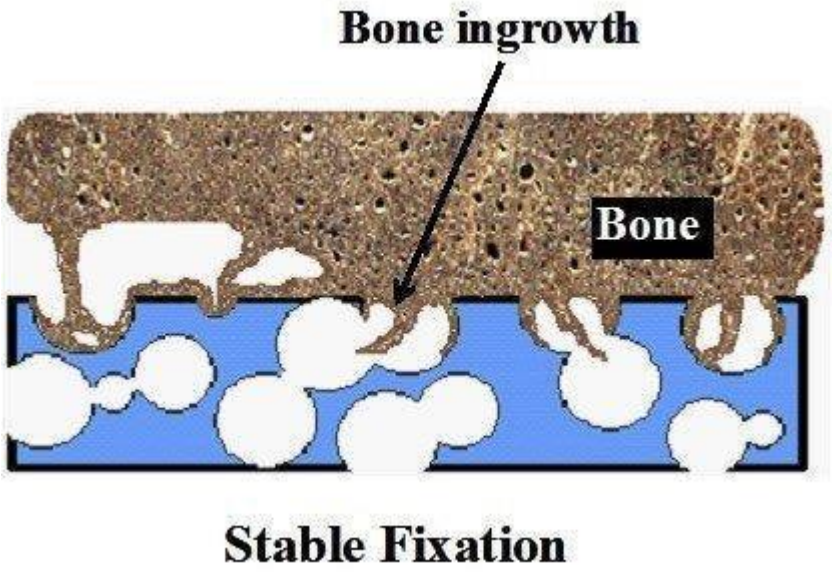
Hydroxyapatite (HAP) Coating: **Hydroxyapatite (HAP)** is a ceramic material whose **chemical composition closely resembles that of natural bone mineral**. Coating the implant with hydroxyapatite encourages **osteointegration** by promoting the growth of new bone tissue on the surface of the implant. The coating helps to accelerate and enhance the bond between the implant and the bone.

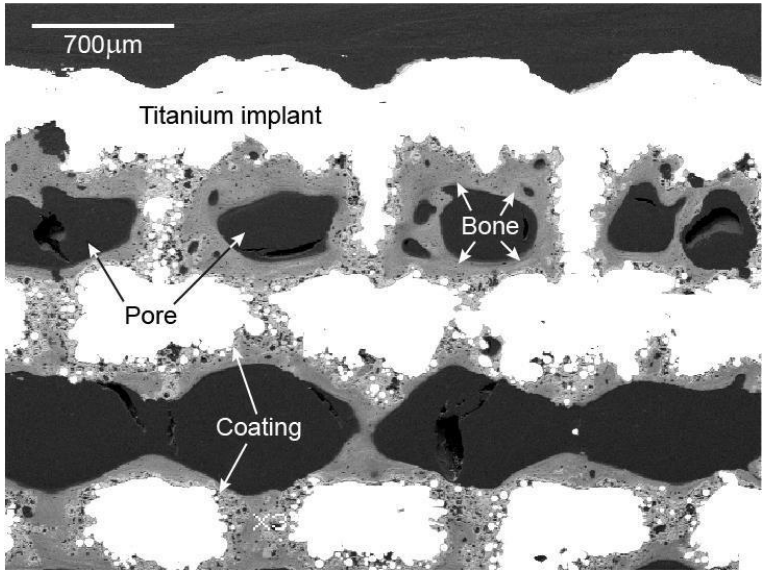


Alternative Fixation: Using porous materials

An alternative to using bone cement is to design implants with **porous surfaces**. These **porous structures** allow **bone ingrowth**, where the bone tissue grows into the pores of the implant, creating a strong **biological fixation**. This method is called **cementless fixation**







Materials selection for acetabular cup and femoral head

They rub against each other. So you have to select accordingly.

Each pairing has specific properties related to wear, longevity, and biocompatibility

Metal-Metal:Used in some older hip replacements but less common now due to concerns about metal ion release and wear particles, which can cause **adverse tissue reactions**.

Ceramic-Ceramic:Known for **excellent wear resistance** and biocompatibility. **Ceramic-on-ceramic** pairings reduce the risk of wear particles but more prone to **fractures.**

•**Metal-UHMWPE (Ultra-High Molecular Weight Polyethylene)**:The most common combination for hip replacements due to good durability and **low cost**. UHMWPE is a **high-wear-resistant polymer**, but over time, it may generate wear debris that can lead to osteolysis.

•**Ceramic (Alumina-Zirconia) - UHMWPE**:This combination is considered one of the **best solutions for wear resistance** because **ceramic femoral heads** paired with **UHMWPE acetabular cups** reduce wear rates significantly compared to metal-on-polyethylene.

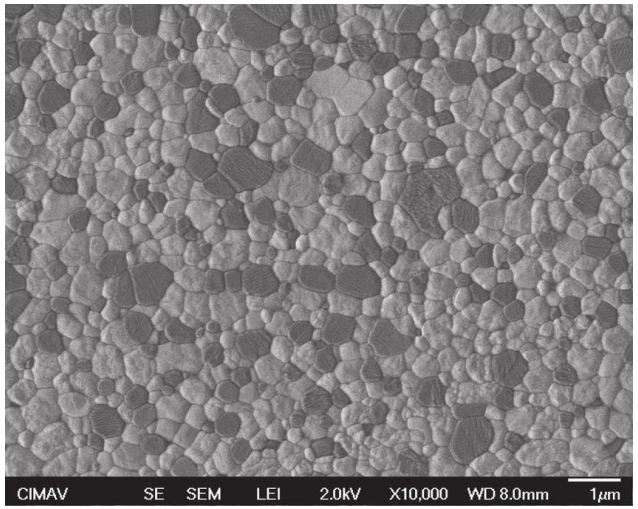
Materials selection for acetabular cup and femoral head

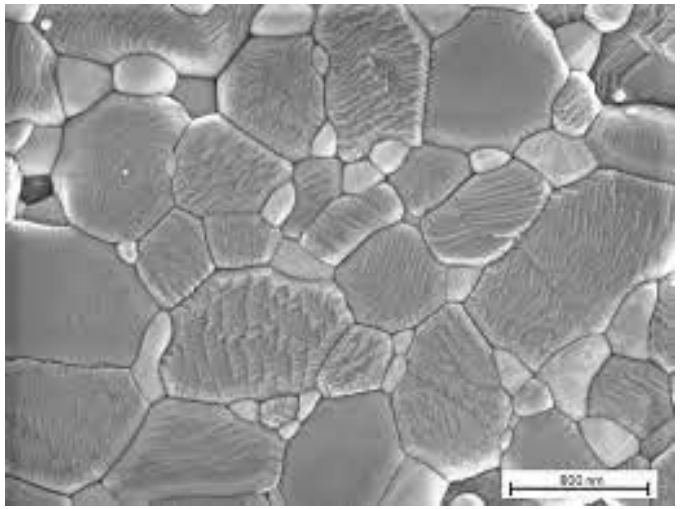
Metal-on-metal implants were first developed in the 1960s but have since been greatly improved. Cobalt chrome is a popular choice, although some studies have found that metal-on-metal implants can cause elevated levels of the metal ions in urine and the bloodstream. This indicates that wear produces particles that enter the body, and which may have an adverse effect. This is particularly a problem for people with poor kidney function.

Studies have shown that increasing the cross-linking in the polyethylene significantly reduces wear, leading to more durable acetabular components, thus increasing the lifetime of an implant

Materials selection for acetabular cup and femoral head

Ceramic-UHMWPE success:
These improved wear resistance properties are dependent on the small, homogeneous grain size in the ceramic, so the microstructure must be carefully controlled during the manufacturing process.





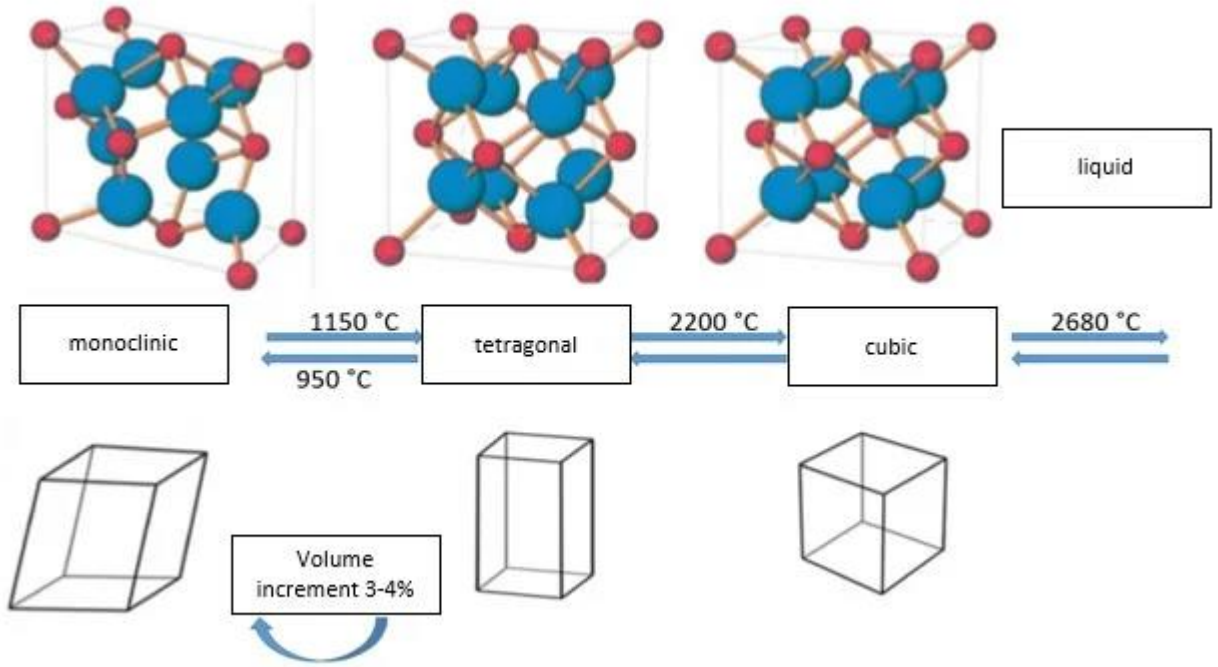
Materials selection for acetabular cup and femoral head

How to impede crack formation and failure in ceramics?

- **Zirconia-Toughened Alumina (ZTA)**: One of the most successful replacements for zirconia is ZTA, which is a composite material that combines alumina (Al_2O_3) with small amounts of zirconia. ZTA benefits from the high wear resistance of alumina and the increased toughness provided by zirconia. This material has better long-term stability and lower risks of phase transformation compared to pure zirconia.
- **Alumina Ceramics**: Pure **alumina** (Al_2O_3) ceramics have long been used in femoral heads due to their excellent wear resistance and bioinertness. Modern alumina is often preferred over zirconia because it avoids the phase transformation issues seen in zirconia.

Transformation toughening in ZTA

Zirconia is a polymorphic material



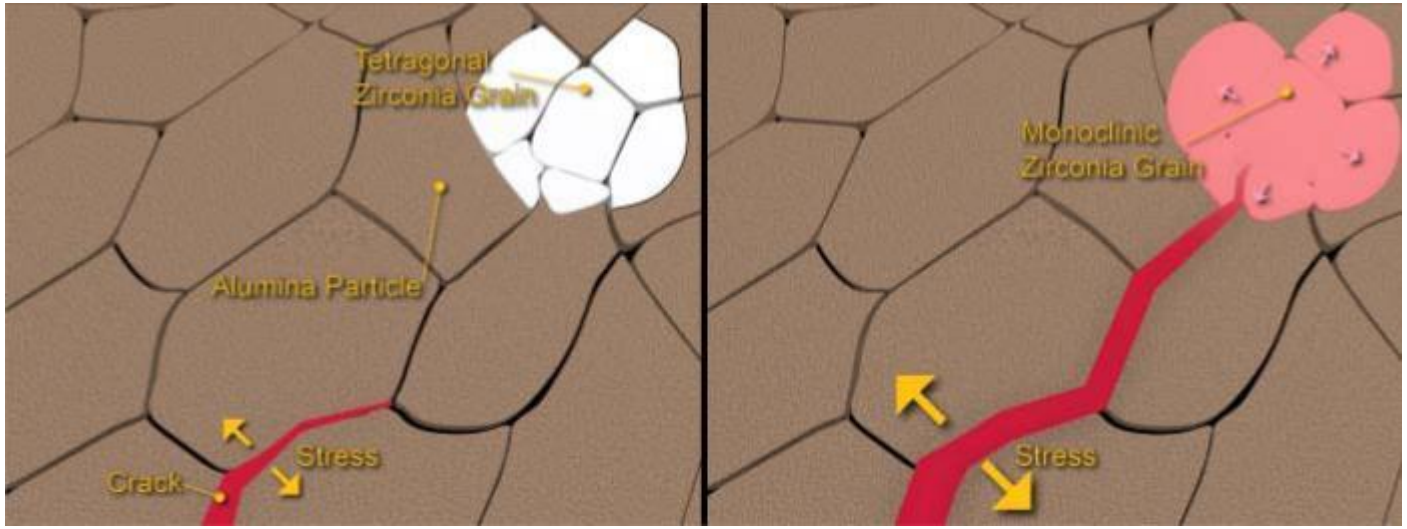
The cubic and tetragonal phases of zirconia can be increased by adding dopants like **Y_2O_3 (yttria)**.

- **7 mol% Y_2O_3**: produces **cubic zirconia**
- **2-6 mol% Y_2O_3**: results in **partially stabilized zirconia (PSZ)**

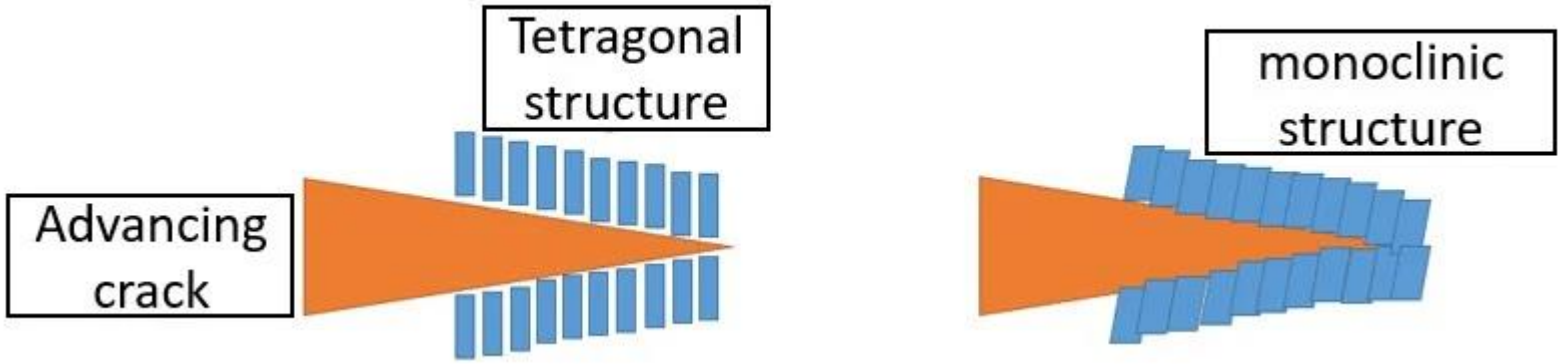
Approximately **10% of the tetragonal phase** is present within a cubic matrix.

phase transformation (tetragonal to monoclinic) helps improve fracture toughness

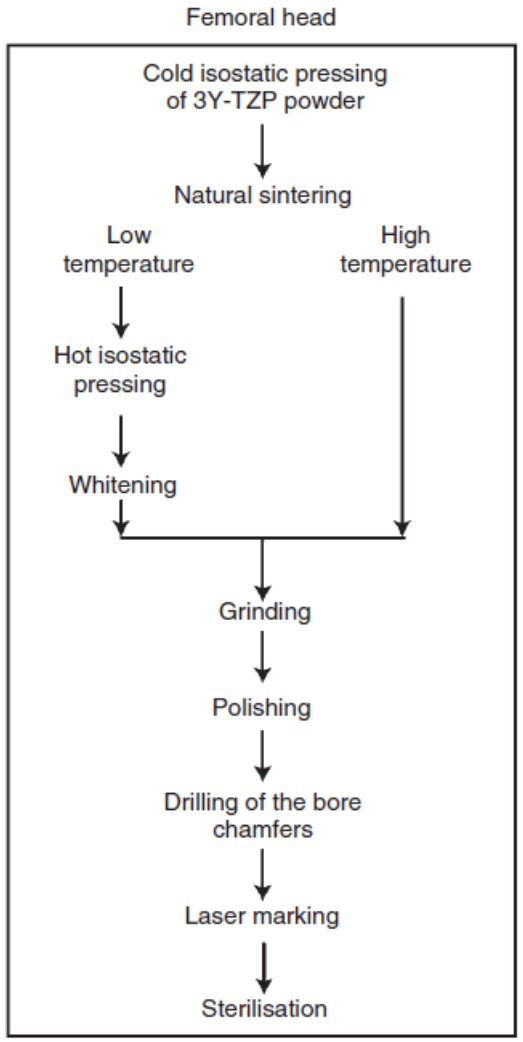
Transformation toughening in ZTA



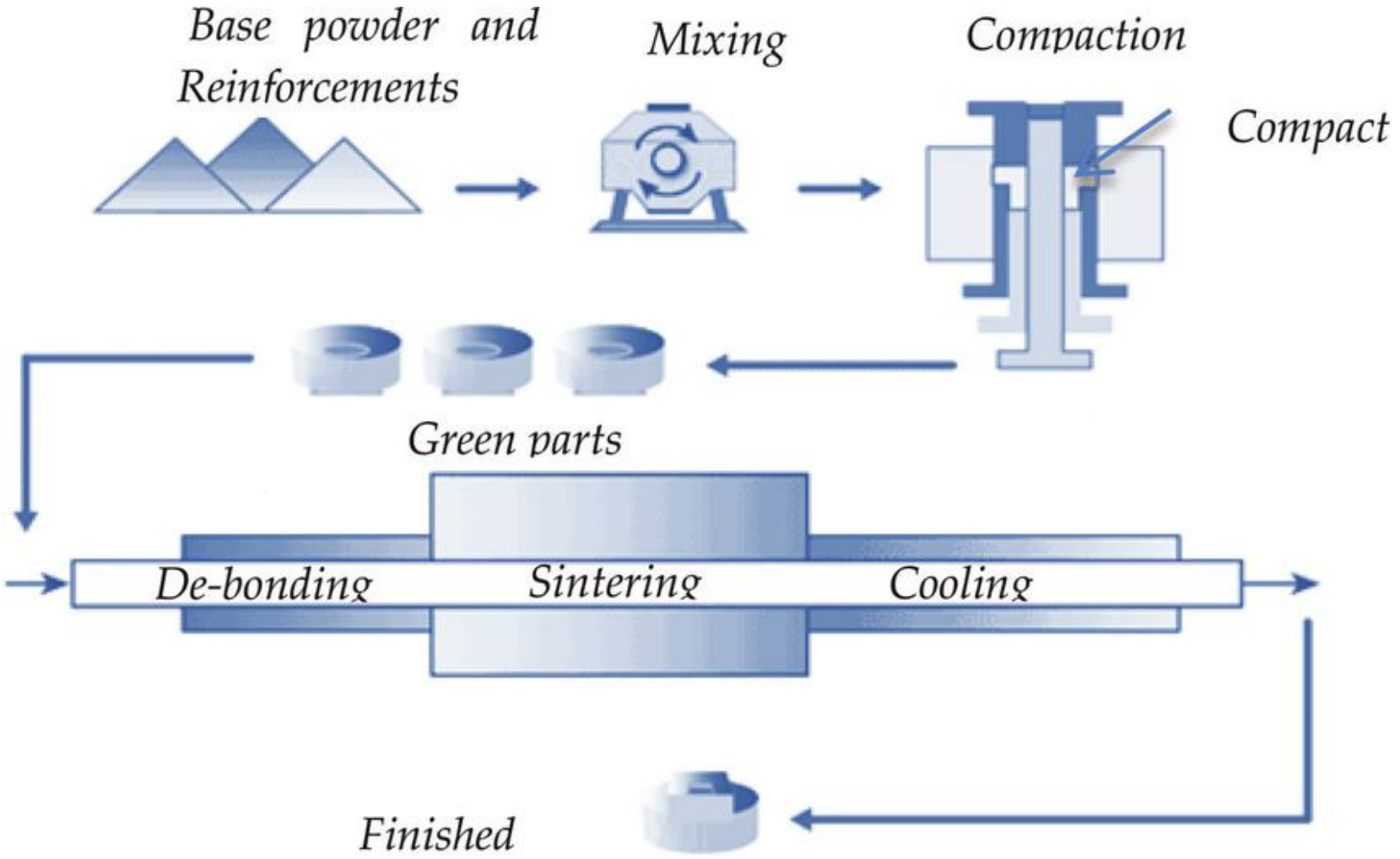
Example of phase transformation and crack propagation (Kyocera Medical).

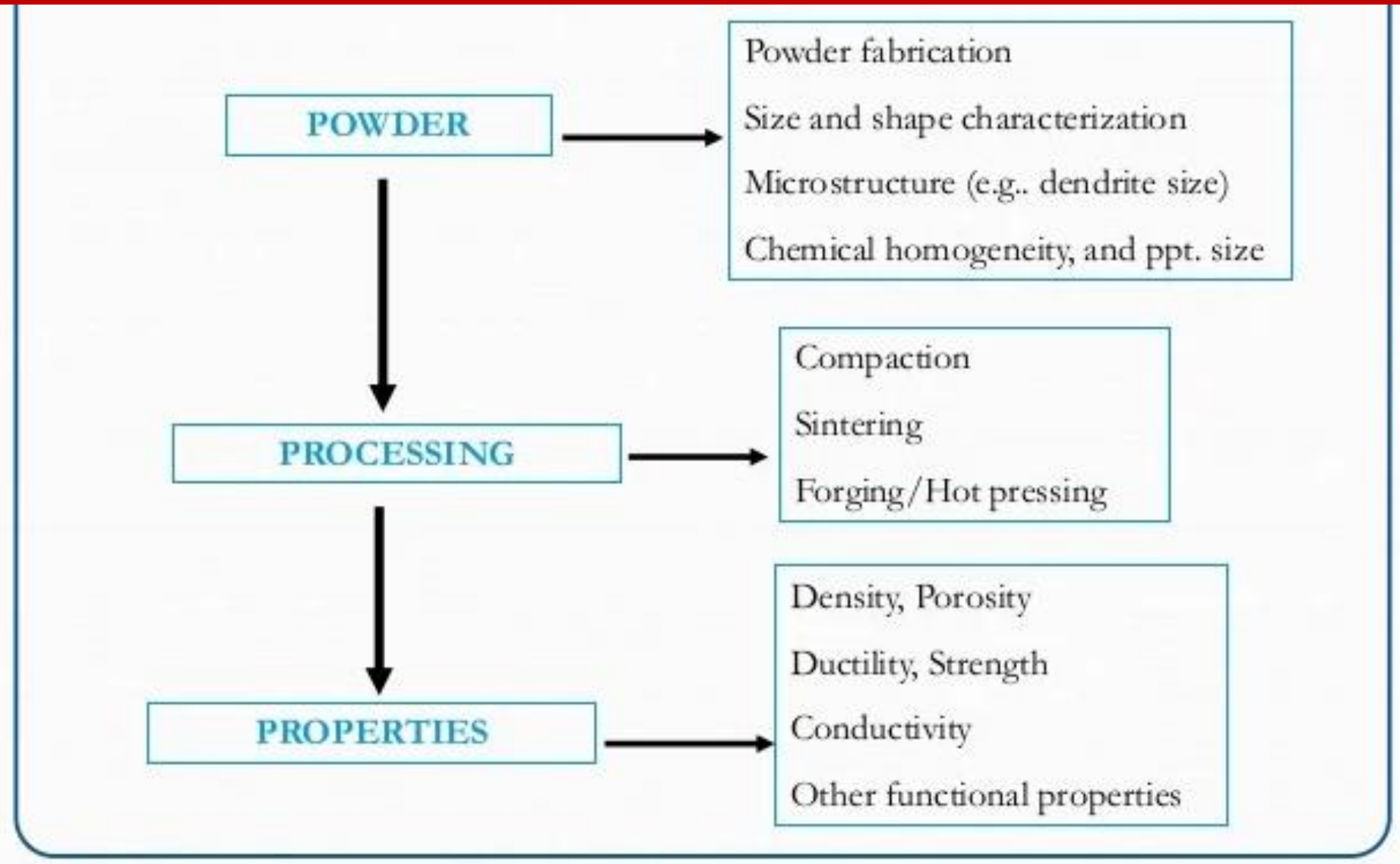


Production method



11.2 Synopsis of the process of zirconia femoral heads.


Base powder and Reinforcements
Mixing
Compaction
Compact
Green parts
De-bonding
Sintering
Cooling
Finished

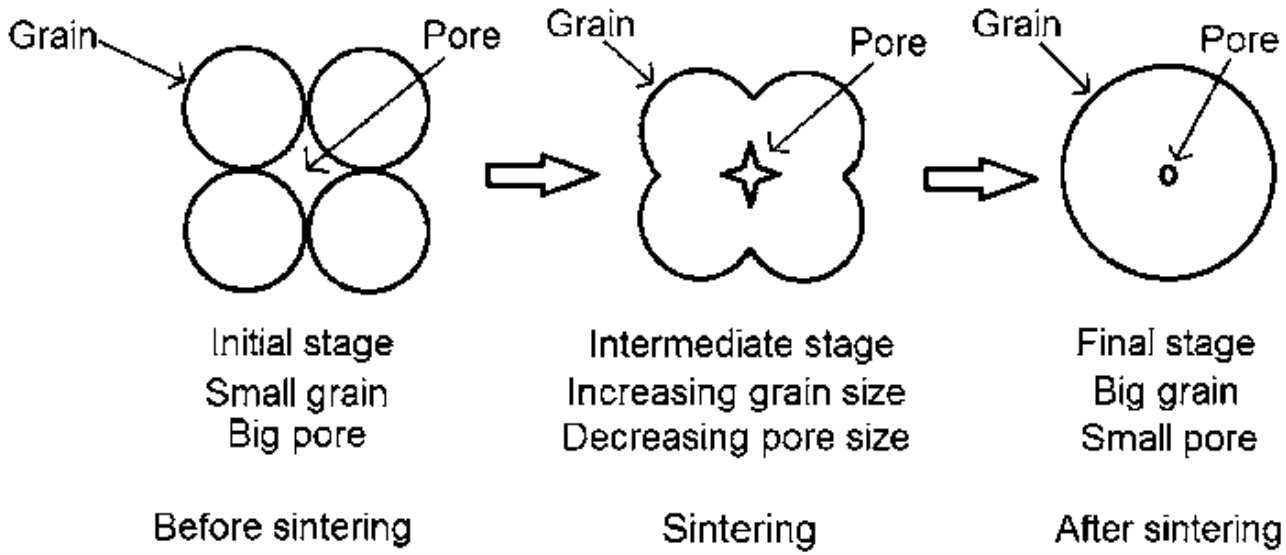

POWDER
Powder fabrication
Size and shape characterization
Microstructure (e.g.. dendrite size)
Chemical homogeneity, and ppt. size
PROCESSING
Compaction
Sintering
Forging/Hot pressing
PROPERTIES
Density, Porosity
Ductility, Strength
Conductivity
Other functional properties

SINTERING

Sintering

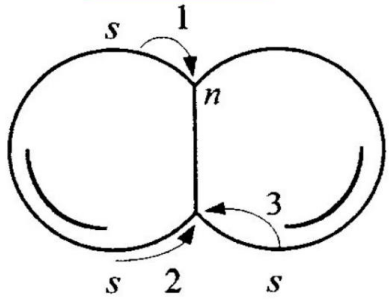
- *Controlled Atmospheres*
- *Sinter/Hardening*
- *Vacuum Sintering*
- *Dimensional Changes*
- *Activated Sintering*
- *Liquid Phase Sintering*

$T_{sintering}$: 85 % of $T_{melting}$



Mass Transport Mechanisms Leading to Densification vs. Coarsening

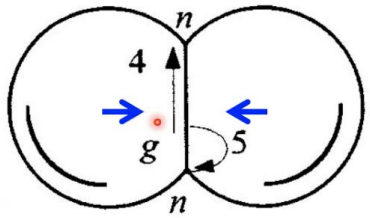
Mechanisms leading to COARSENING



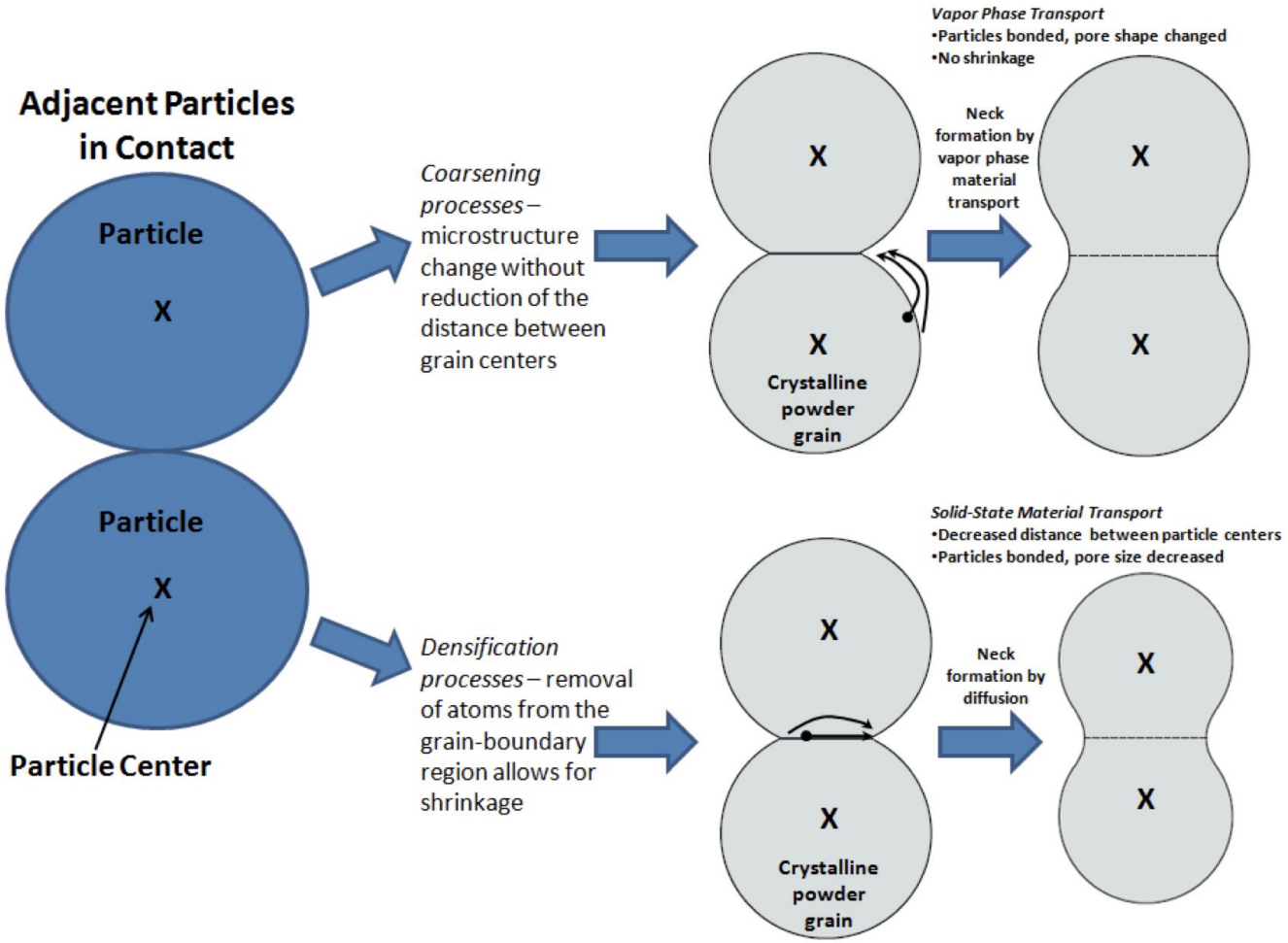
1. Evaporation-condensation from surface to "neck"
2. Surface diffusion to "neck"
3. Lattice diffusion from surface to "neck"

M W Barsoum (2003), Fundamentals of Ceramics, p. 309-315

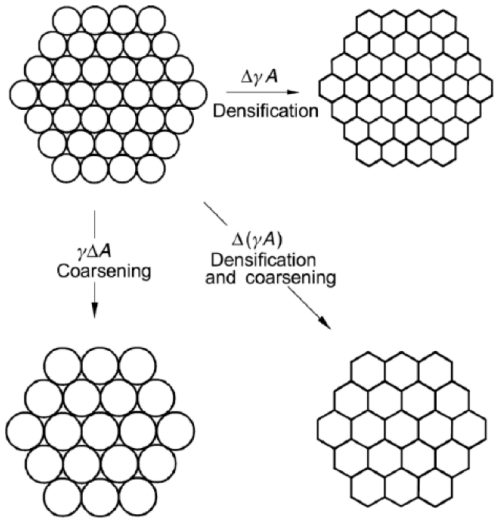
Mechanisms leading to DENSIFICATION



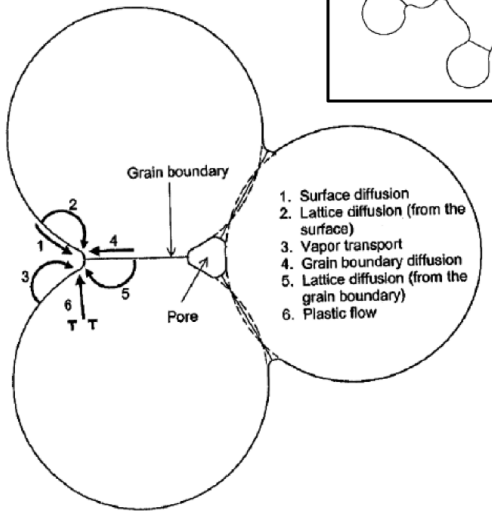
4. Grain boundary (GB) diffusion from GB center to "neck"
5. Lattice diffusion from GB center to "neck"
6. Plastic/viscous (for glass) flow to "neck"


Adjacent Particles in Contact
Particle
X
Particle
X
Particle Center
Coarsening processes – microstructure change without reduction of the distance between grain centers
Densification processes – removal of atoms from the grain-boundary region allows for shrinkage
Vapor Phase Transport
•Particles bonded, pore shape changed
•No shrinkage
Neck formation by vapor phase material transport
Crystalline powder grain
Solid-State Material Transport
•Decreased distance between particle centers
•Particles bonded, pore size decreased
Neck formation by diffusion
Crystalline powder grain

Sintering

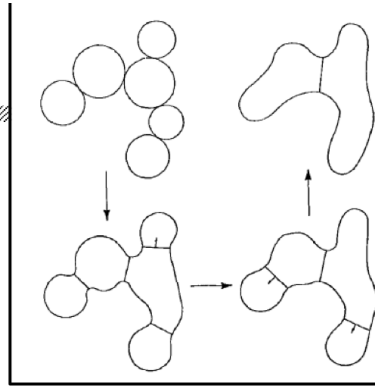


(a) Sintering basic phenomena of densification and coarsening under the sintering driving force.

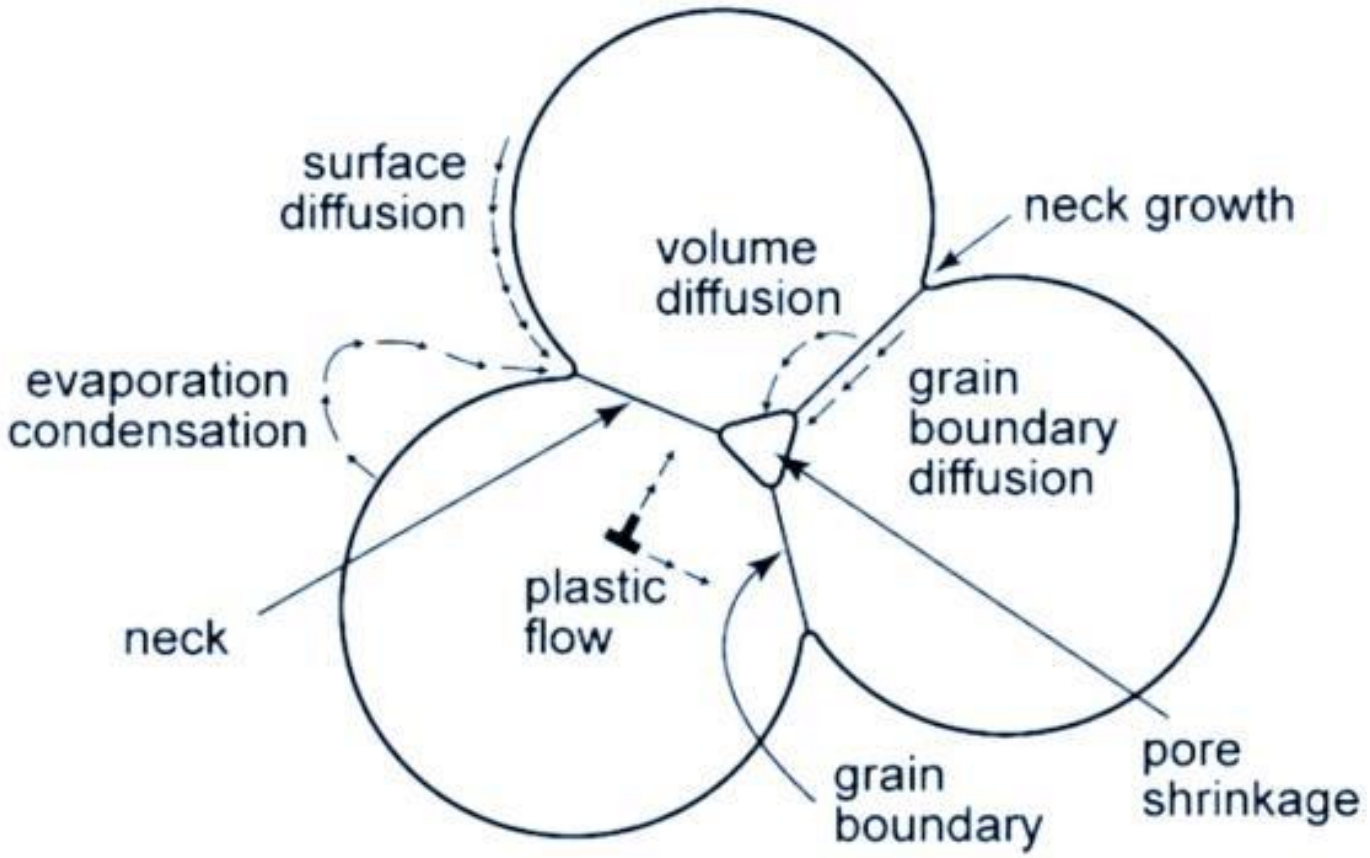


1. Surface diffusion
2. Lattice diffusion (from the surface)
3. Vapor transport
4. Grain boundary diffusion
5. Lattice diffusion (from the grain boundary)
6. Plastic flow

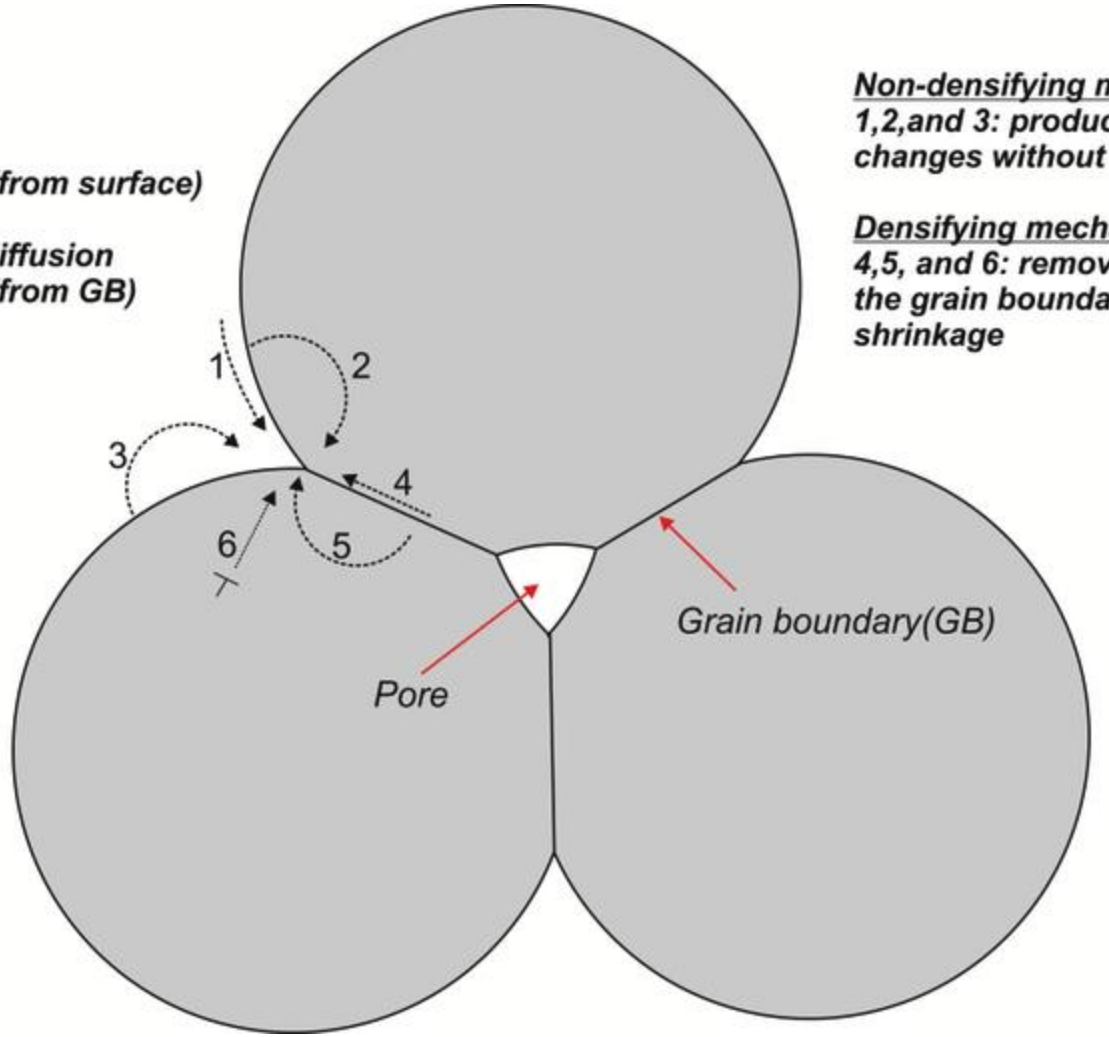
(b) Basic sintering mechanisms.



The smallest the particle size (larger surface area) the **higher the sintering driving force**

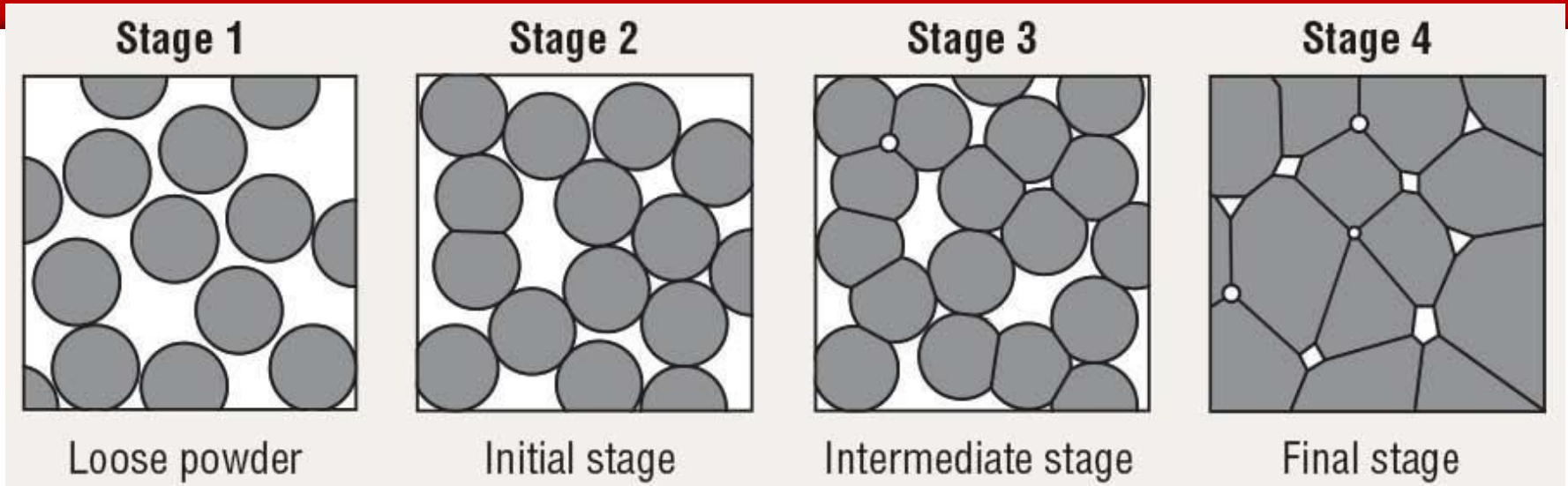

surface
diffusion
volume
diffusion
neck growth
evaporation
condensation
grain
boundary
diffusion
plastic
flow
neck
grain
boundary
pore
shrinkage

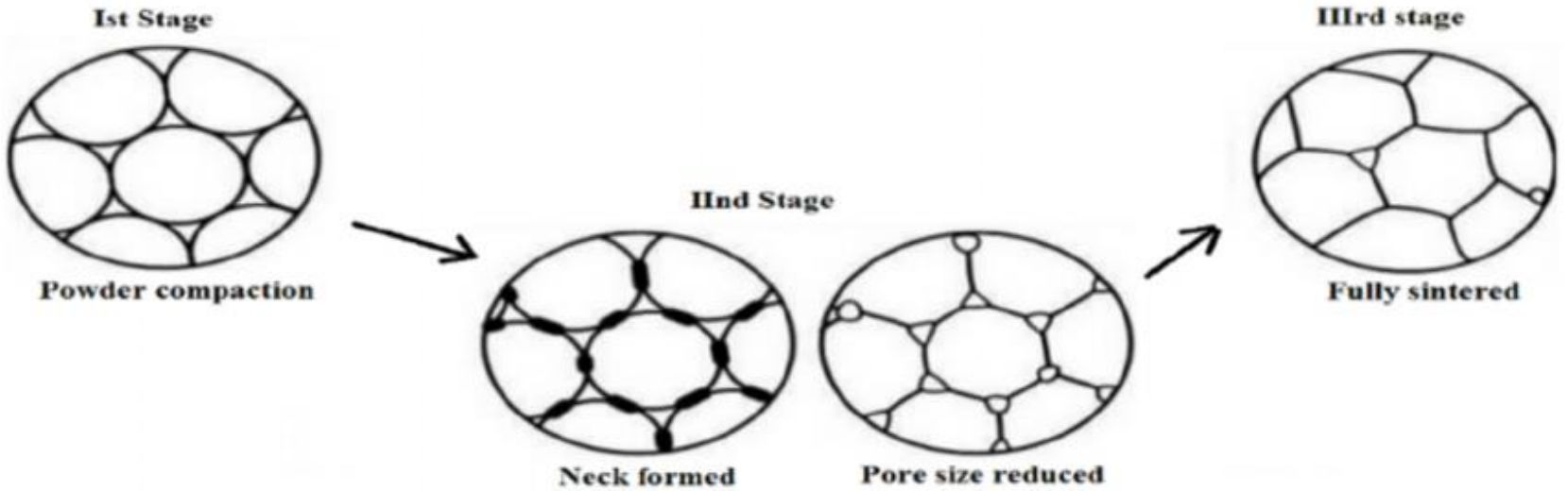
Mechanisms
1. *Surface diffusion*
2. *Lattice diffusion (from surface)*
3. *Vapor transport*
4. *Grain boundary diffusion*
5. *Lattice diffusion (from GB)*
6. *Plastic flow*



Non-densifying mechanisms:
1,2,and 3: produce microstructural changes without causing shrinkage

Densifying mechanisms:
4,5, and 6: remove material from the grain boundary region causing shrinkage


Stage 1
Stage 2
Stage 3
Stage 4
Loose powder
Initial stage
Intermediate stage
Final stage


Ist Stage
Powder compaction
IInd Stage
Neck formed
Pore size reduced
IIIrd stage
Fully sintered

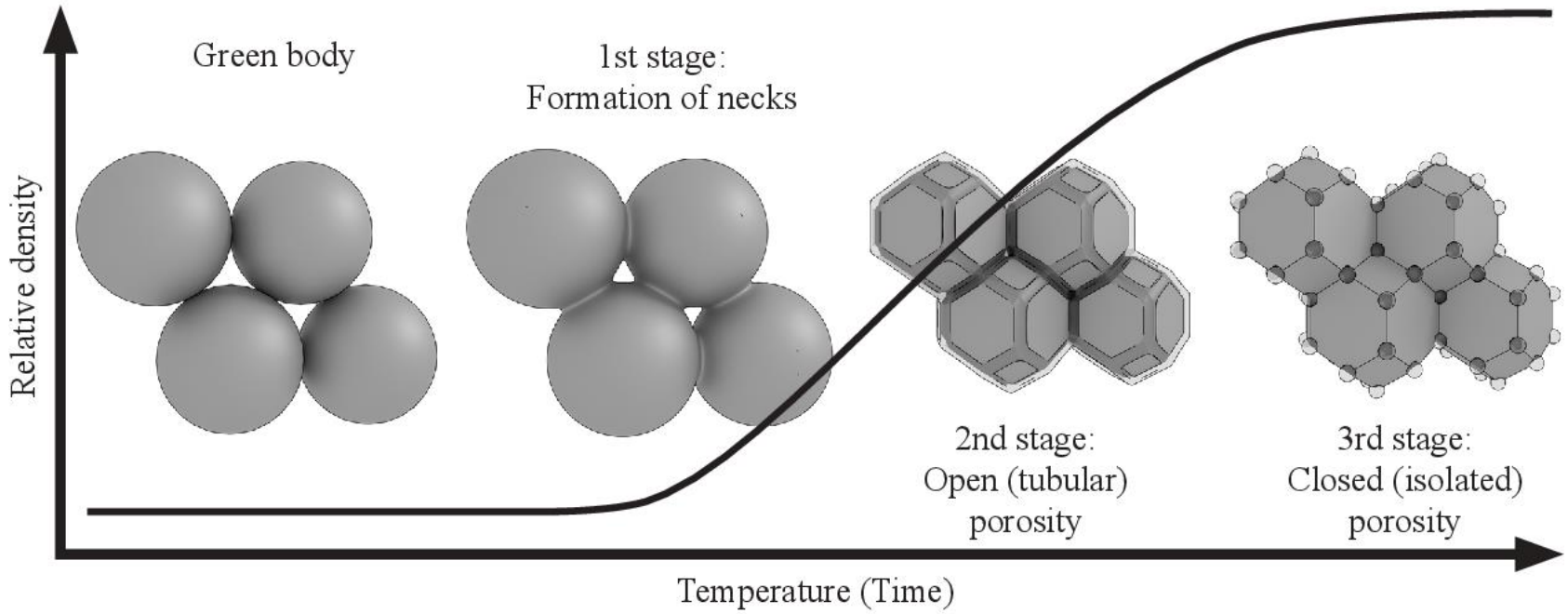
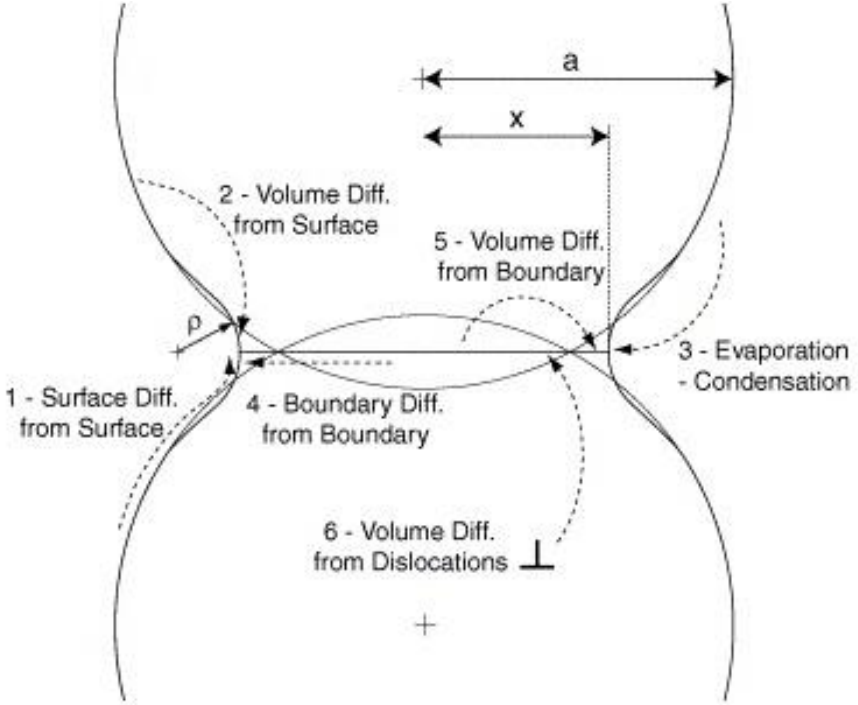


Figure 6: Stages of sintering.

6 SINTERING MECHANISMS

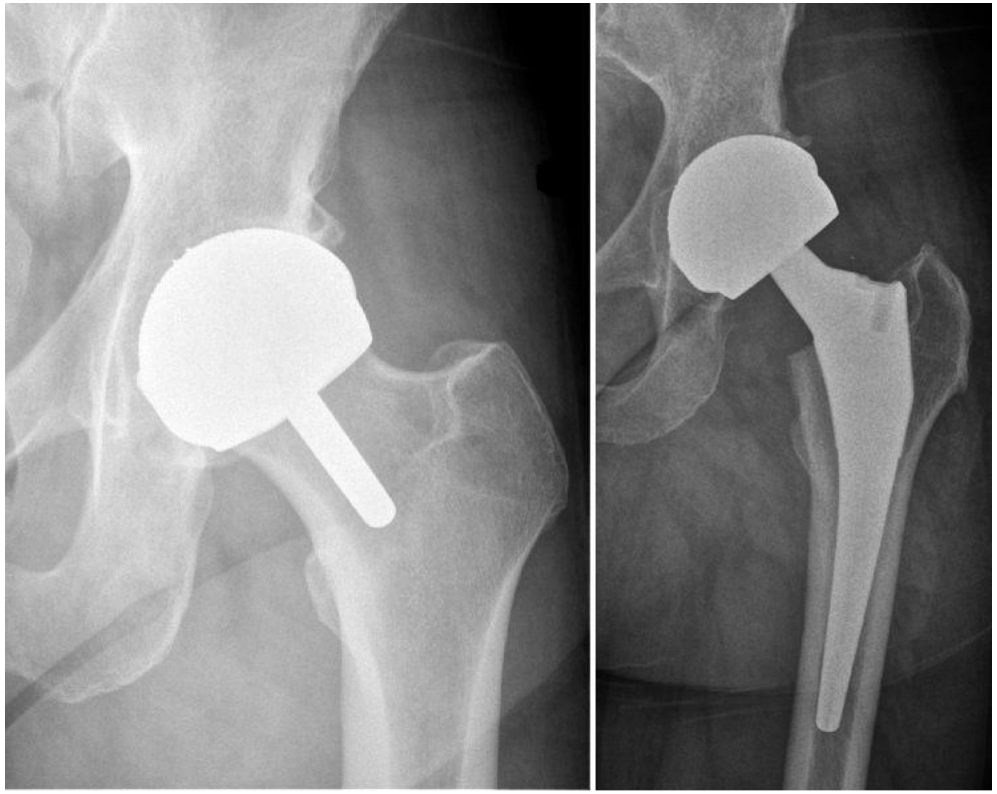


The six different sintering mechanisms considered by Ashby in the construction of sintering diagrams

The relationship between strength and grain size in ceramics

•**Less Surface Roughness**: Smaller grains can result in a smoother surface, which helps reduce wear. Smoother surfaces create less friction and minimize the chances of abrasive effects causing wear.
•**Higher Hardness**: Ceramic materials are known for their high hardness, and smaller grains often lead to increased hardness. Greater hardness can improve the material's resistance to wear, as it becomes less prone to deformation or abrasion.
•**Improved Durability**: While ceramics are known for their hardness, they can also be brittle. Smaller grains can enhance the durability or resistance to fracture of ceramic materials, making them more reliable in demanding applications.
•**Reduced Crack Propagation**: Smaller grains can help prevent the spread of cracks or defects within the material. This is crucial for avoiding sudden failure and maintaining wear resistance over time.
•**Homogeneity**: Smaller grains contribute to a more homogeneous microstructure, which can result in consistent wear properties throughout the material.

Other option for hip replacement: hip resurfacing



Hip Resurfacing

Traditional Total Hip Replacement

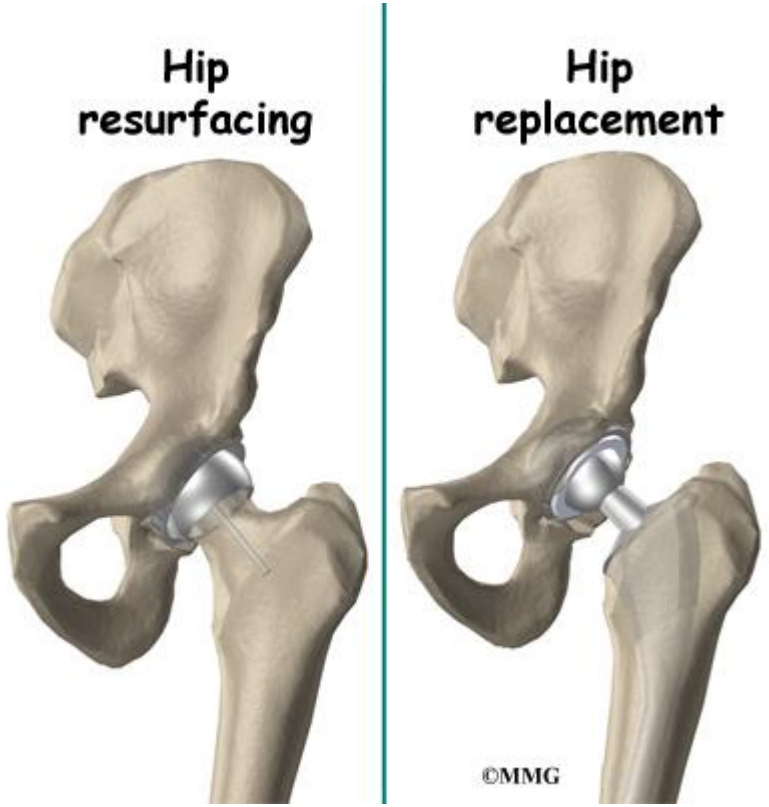
In a traditional total hip replacement surgery, both the head of the thigh bone (femoral head) and the damaged socket (acetabulum) are removed and replaced with components made of metal, plastic, or ceramic.

In **hip resurfacing**, however, the femoral head is not removed; instead, it is covered with a smooth metal cap. The damaged bone and cartilage inside the socket are removed, and they are replaced with a metal shell, similar to that used in traditional total hip replacement surgery.

Metal-on-metal hip resurfacing often uses larger femoral heads. A larger head size can improve joint stability, reduce the risk of dislocation, and enhance joint function.

Hip resurfacing





Life span

Standard total hip replacement (THR) designs offer excellent pain relief and predictable outcomes, typically providing a 10-year survival rate of 93% to 97% and a 25-year implant survival rate of 60% to 80%.

Bibliography

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10.1016/j.jor.2020.12.036

10.1016/j.jmbbm.2013.03.022